Notes for MSQE students

Essential Mathematics for Economic Analysis[1]

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[1]Please do not circulate these notes.

Contents

1 Set Theory and Logic (on 07/13)

We start with set theory and establish the basic terminology and notation. We shall also discuss some points of elementary logic.

1.1 Sets, Unions, and Intersections

A *set* is a collection of elements (intuitively it is another way of saying category or label). For instance, a set could be sausages. It's very general and encompasses many different things. However, all these things have one thing in common; the fact that they are sausages. One such thing would be a spinach and feta chicken sausage and we call this an *element*. As economists, we deal with very large sets with many elements (i.e. the consumption set which includes all possible consumption bundles). Thus, it is very useful to use general notation. The norm is to denote sets by capital letters such as A, B, X, and Y, and elements of these sets by lowercase letters such as a, b, x, and y. In our sausage example, S is the set of all sausages. If an element s belongs to the set of sausages, we express this fact by:

$$s \in S.$$

If s does not belong to S, we express this fact by writing

$$s \notin S.$$

We say that A is a *subset* of B if every element of A is also an element of B; and we express this by writing

$$A \subset B.$$ [2]

To refer back to my example; let C be the set of chicken sausages, then $C \subset S$.

What is the proper way of specifying a set? If the set has only a few elements, we can just list the objects in the set. In words "A is the set consisting of the elements a, b, and c." In symbols, this statement is

$$A = \{a, b, c\},$$

where braces are used to enclose the list of elements. However, in most cases (like our sausage example) you can imagine there to be too many elements to realistically write out.

[2] Note, however, that in general the definition for subset does not require A to be different from B; in fact, if $A = B$, then both $A \subset B$ and $B \subset A$. For our purposes, this technical nuance won't apply.

For instance, one might take the set of all even integers. This would be written as

$$B = \{x \mid x \text{ is an even integer}\}.$$

Here the braces stand for the words "the set of," and the vertical bar stands for the words "such that." The equation is read "B is the set of all x such that x is an even integer."

Examples: Empty "Null" set = $\emptyset$. Nonnegative real numbers = $\mathbb{R}_+ \equiv \{x \in \mathbb{R} : x \geq 0\}$.

1.1.1 The Union of Sets

The *union* of two sets A and B, denoted $A \cup B$, is the set which consists of all elements which are either in A or in B (or both):[3]

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

1.1.2 The Intersection of Sets

The *intersection* of two sets A and B, denoted $A \cap B$, is the set which consists of all elements which belong to both A and B:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

Note, if A is the set of Chicken and B is the set of Sausages, then $A \cap B$ would be the set of Chicken Sausages.

1.1.3 The Complement of Sets

If it is clear that all sets under discussion are subsets of some universal set U, then $U - A = A^c$ is said to be the complement of A (in U).

1.2 Numbers

The most basic numbers are called **natural numbers**. The set of natural numbers is usually denoted by $\mathbb{N}$:

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

[3] Mathematicians have agreed that they will use the word "or" to always include "or both". If one means P or Q, but not both," then one has to include the phrase "but not both" explicitly.

The next most basic numbers are called **integers**. The set of integers is usually denoted by

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

The set $\mathbb{Q}$ is called the set of **rational numbers**, since it is formed by taking the ratios of integers:

$$\mathbb{Q} \equiv \left\{\frac{a}{b} : \; a, b \in \mathbb{Z}; \; b \neq 0\right\}$$

Numbers that cannot be written as ratios or quotients of integers are called **irrational numbers**. The set of rational and irrational numbers is called the set of **real numbers** and is denoted with $\mathbb{R}$.

Definition 1. *An integer n is called an **even number** if there is an integer m such that $n = 2m$. An integer that is not even is called an **odd integer**.*

1.2.1 Properties of Addition and Multiplication

1. **(Closure)** If a and b are in $\mathbb{R}$, so are $a + b$ and $a \cdot b$.

2. **(Commutative)** For any $a, b \in \mathbb{R}$, $a + b = b + a$ and $a \cdot b = b \cdot a$.

3. **(Associative)** If $a, b, c \in \mathbb{R}$, $(a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

4. **(Identity)** There is an element $0 \in \mathbb{R}$ such that $a + 0 = a$ for all $a \in \mathbb{R}$.The is an element $1 \in \mathbb{R}$ such that $a \cdot 1 = a$ for all $a \in \mathbb{R}$.

5. **(Inverse)** for an $a \in \mathbb{R}$, there is an element $b \in \mathbb{R}$ such that $a + b = 0$; such a b is usually written as $-a$. For any nonzero $a \in \mathbb{R}$, there is an element $c \in \mathbb{R}$ such that $a \cdot c = 1$, such a c is usually written as $1/a$.

6. **(Distributive)** For all $a, b, c \in \mathbb{R}$, $a \cdot (b + c) = a \cdot b + a \cdot c$.

1.2.2 Bounds

Definition 2. *Let S be a subset of $\mathbb{R}$ and let $b \in \mathbb{R}$. Then, the number b is called an **upper bound** for S if $a \leq b$ for all $a \in S$; the number b is called a **lower bound** for S if $b \leq a$ for all $a \in S$.*

Definition 3. *If b is an upper bound for S and no element smaller than b is an upper bound for S, then b is called a **least upper bound (lub)** for S. Similarly, if b is a lower bound for S and no number lager than b is a lower bound of S, then b is called a **greatest lower bound (glb)** for S.*

Suppose $S = [0, 1]$. What about $S = (0, 1)$.

1.3 Propositions: Contrapositives and Converses

A proposition is an assertion that is either true or false. Suppose P and Q are two propositions such that whenever P is true, then Q is necessarily true. In this case, we usually write

$$P \Rightarrow Q \tag{1.1}$$

This is read as "P implies Q," or "if P, then Q," or "Q is a consequence of P." It is important to note that $P \Rightarrow Q$ **only** says that if P is true, then Q is also true. It does NOT say anything about the whether P is *not* true; in this case, Q could be either true or false. For example, suppose P is the statement $x > 0$ and Q is the statement that $x^2 > 0$. It is obvious that $P \Rightarrow Q$, but Q can be true even if P is not true.

1.3.1 Contrapositives

Given a statement of the form "if P, then Q," its *contrapositive* is the statement that "if Q is not true, then P is not true." For example, the contrapositive of the statement

$$\text{If } x \text{ is positive, then } x^3 \text{ is positive}$$

is the statement

$$\text{If } x^3 \text{ is not positive, then } x \text{ is not positive.}$$

Note that a statement and its contrapositive are logically equivalent. This fact can come in handy as occasionally it is easier to prove the contrapositive than the actual statement.

1.3.2 Converses

The *converse* of a statement $P \Rightarrow Q$ is the statement that

$$Q \Rightarrow P$$

that is, the statement that "if Q, then P." BE AWARE: there is no logical relationship between a statement and its converse! Thus, if $P \Rightarrow Q$, then it is not necessarily the case that $Q \Rightarrow P$.

If a statement and its converse both hold, we express this by saying that "P if and only if Q," or "P iff Q," and denote this by

$$P \Leftrightarrow Q.$$

1.3.3 Types of Proofs

Constructive proof (direct proof): Assume that P is true, deduce various consequences, and use them to show that Q must also hold.

Contrapositive proof (indirect proof): Assume that Q does *not* hold, then show that P cannot hold either i.e. if P implies Q, then "not Q" implies "not P".

Proof by contradiction: Assume that P is true and assume that Q is *not* true and then derive a logical contradiction (very efficient means of proof but often throws very little light on nature of relation between P and Q - but used quite often in microeconomics).

Proof by induction: Show first that the formula is valid for $n = 1$, second that, if the formula is valid for $n = k$, then it is also valid for $n = k + 1$.

Sample Proofs:

1. Use two methods of proof to prove that

$$-x^2 + 5x - 4 > 0 \Rightarrow x > 0$$

(a) *Direct Proof:* Suppose $-x^2 + 5x - 4 > 0$. Adding $x^2 + 4$ to each side of the inequality gives $5x > x^2 + 4$. Because $x^2 + 4 \geq 4$, for all x, we have $5x > 4$, so $x > 4/5 > 0$.

(b) *Indirect Proof:* Suppose $x \leq 0$. Then $5x \leq 0$ and so $-x^2 + 5x - 4$, as a sum of three nonpositive terms, is ≤ 0.

2. Prove by induction that, for all positive integers n,

$$3 + 3^2 + 3^3 + 3^4 + \cdots + 3^n = \frac{1}{2}\left(3^{n+1} - 3\right)$$

Solution: For $n = 1$, both sides are 3. Suppose the expression is true for $n = k$. Then

$$3 + 3^2 + 3^3 + 3^4 + \cdots + 3^k + 3^{k+1} = \frac{1}{2}\left(3^{n+1} - 3\right) + 3^{k+1} = \frac{1}{2}\left(3^{k+2} - 3\right)$$

which is the expression for $n = k + 1$. Thus, by induction, the expression is true for all n.

2 Limits and Open Sets (on 07/13)

2.1 Sequences of Real Numbers

Definition 4. *The* ***Natural numbers*** *– also called the* ***positive integers*** *– are just the usual counting numbers: 1, 2, 3,. . . .*

Definition 5. *A* ***sequence*** *of real numbers is an assignment of a real number to each natural number usually written as* $\{x_1, x_2, x_3, \ldots, x_n, \ldots\}$ *where* x_1 *is the real number assigned to the natural number 1 – generally written as* $\{x_n\}_{n=1}^{\infty}$.

Some examples are

$$\{1, 2, 3, 4, \ldots\} \tag{1.2}$$

$$\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\} \tag{1.3}$$

$$\left\{1, \frac{1}{2}, 4, \frac{1}{8}, 16, \ldots\right\} \tag{1.4}$$

Limit of a Sequence

There are basically three kinds of sequences:

1. sequences like (1.2) in which the entries increase without bound;
2. sequences like (1.3) in which the entries get closer and closer and stay close to some limiting value; and
3. sequences like (1.4) in which neither behavior occurs so that the entries jump back and forth on the number line.

Definition 6. *Let* $\{x_n\}_{n=1}^{\infty}$ *be a sequence of real numbers and let* r *be a real number. We say that* r *is the* ***limit*** *of this sequence if for any (small) positive number* ε*, there is a positive integer* N *such that for all* $n \geq N$*,* x_n *is in the* ϵ*-interval about* r*; that is*

$$|x_n - r| < \varepsilon.$$

*In this case, we say that the sequence **converges to** r and we write*

$$\lim x_n = r \text{ or } \lim_{n\to\infty} x_n = r \text{ or simply } x_n \to r.$$

Theorem 1. *A sequence can have at most one limit.*

Theorem 2. *Let $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ be sequences with limits x and y, respectively. Then the sequence $\{x_n + y_n\}_{n=1}^{\infty}$ converges to the limit $x + y$.*

Theorem 3. *Let $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ be sequences with limits x and y, respectively. Then the sequence of products $\{x_n y_n\}_{n=1}^{\infty}$ converges to the limit xy.*

2.2 Sequences in $\mathbb{R}^m$

Definition 7. *The **Euclidean metric** is a distance measure defined as*

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_m - y_m)^2}$$

Definition 8. *Let $\boldsymbol{r}$ be a vector in $\mathbb{R}^m$ and let ε be a positive number. The ε-**ball about** $\boldsymbol{r}$ is*

$$B_\varepsilon(\mathbf{r}) \equiv \{\mathbf{x} \in \mathbb{R}^m : \|\mathbf{x} - \mathbf{r}\| < \varepsilon\}.$$

Definition 9. *A sequence of vectors $\{\mathbf{x}_n\}_{n=1}^{\infty}$ is said to **converge** to the vector $\mathbf{y}$ if for any choice of a positive real number ε, there is an integer N such that for all $n \geq N$, $\mathbf{x}_n \in B_\varepsilon(\mathbf{x})$; that is*

$$d(\mathbf{x}_n, \mathbf{y}) = \|\mathbf{x}_n - \mathbf{y}\| < \varepsilon.$$

2.3 Open and Closed Sets

Definition 10. *A set S in $\mathbb{R}^m$ is **open** if for each $\mathbf{x} \in S$, there exists an open ε-ball around $\mathbf{x}$ completely contained in S:*

$$\mathbf{x} \in S \Rightarrow \exists \text{ an } \varepsilon > 0 \text{ such that } B_\varepsilon(\mathbf{x}) \subset S.$$

Example: The interval

$$(0, 1) = \{x \in \mathbb{R} : 0 < x < 1\}$$

is an open set.

Definition 11. *A set S in $\mathbb{R}^m$ is* ***closed*** *if, whenever $\{\mathbf{x}_n\}_{n=1}^{\infty}$ is a convergent sequence completely contained in S, its limit is also contained in S.*

Example: The interval

$$[0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$$

is an closed set.

Theorem 4. *A set S in $\mathbb{R}^m$ is closed if and only if its complement, $S^c = \mathbb{R}^m - S$, is open.*

Definition 12. *A point $\mathbf{x}$ is in the* ***boundary*** *of a set S is every open ball about $\mathbf{x}$ contains both points in S and points in the complement of S.*

3 Linear Algebra (on 07/13 and 07/14)

Example: Here is the following system of equations for x_1, x_2, and x_3.

$$\begin{aligned} 6x_1 + 3x_2 + x_3 &= 22 && (2.1) \\ x_1 + 4x_2 - 2x_3 &= 12 && (2.2) \\ 4x_1 - x_2 + 5x_3 &= 10 && (2.3) \end{aligned}$$

The usual questions of interest when we face such a system consist of:
1. Does a solution exist?
2. If a solution exists, is the solution unique?
3. What is the solution?
In order to answer 1 and 2, as so far we do not know anything about matrix algebra, we need to find the solution to the above system of equations.

In order to solve the system, let us write x_1 as a function of both x_2 and x_3 in equation 2.2. Doing so we get $x_1 = 12 - 4x_2 + 2x_3$. Replacing that expression into equations 2.1 and 2.3, we obtain

$$\begin{aligned} -21x_2 + 13x_3 &= -50 && (2.4) \\ -17x_2 + 13x_3 &= -38 && (2.5) \end{aligned}$$

We have transformed the original system of 3 equations with 3 unknowns into a system of two equations with two unknowns. This system is very easy to solve. We express x_3 as a function of x_2. We get

$$-50 + 21x_2 = -38 + 17x_2 \tag{2.6}$$

From this, we have that $x_2 = 3$. Replacing that into any of the two equations above, we get that $x_3 = 1$. Finally the value of x_1 is obtained $x_1 = 2$. A solution exists and is unique. The solution can be expressed as follows $\{x_1, x_2, x_3\} = \{2, 3, 1\}$.

This is a very easy example as we only have 3 equations and 3 unknowns. Sometimes we have more than 3 equations and more than 3 unknowns and it can be very long and cumbersome to solve the system. Even, sometimes we are not interested in the exact value of the solution; rather we want to know whether a solution exists and whether that solution is unique. This can be achieved, quite easily, using matrix algebra. The only caveat is that the system must be of *linear* equations...hence the term "Linear" Algebra. Another benefit to Linear (or matrix) algebra is it makes notation much more compact.

For instance, take a system of m linear equations with n unknowns:

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= b_2 \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= b_m
\end{aligned}$$

This system can be written in matrix form as follows

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Or in a more compact notation, $\mathbf{Ax} = \mathbf{b}$. Note that boldfaced uppercase letters generally refer to matrices and boldfaced lowercase letters refer to vectors.[4] The m equations create m **rows** (equations) and the n variables (unknowns) n **columns**. The $m \times n$ matrix $\mathbf{A}$ is known as the **coefficient matrix**, the $n \times 1$ matrix $\mathbf{x}$ and the $m \times 1$ matrix $\mathbf{b}$ are known as column **vectors**.

Example: Rewrite the above example in matrix algebra form.

[4] Remember that a vector is a special kind of matrix.

$$\begin{bmatrix} 6 & 3 & 1 \\ 1 & 4 & -2 \\ 4 & -1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix}$$

3.1 Basic Matrix Algebra

A matrix is an ordered collection of numbers (or variables).

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

The number a_{ij} refers to the *element* in row i, column j. The above matrix is often expressed as,

$$[a_{ij}]_{m\times n} \text{ or } \mathbf{A}_{m\times n}$$

A matrix which has the same number of rows as columns, $m = n$, is a **square matrix**.

3.1.1 Addition, Subtraction and multiplication by a scalar

Two matrices $\mathbf{A} = [a_{ij}]_{m\times n}$ and $\mathbf{B} = [b_{ij}]_{m'\times n'}$ are said to be equal if and only if they have the same dimension, i.e. $m = m',\ n = n'$, **and** have identical elements in the corresponding locations in the array.

$\rightarrow$ To add (subtract) to matrices simply add(subtract) the respective elements together. **NOTE:** in order to add or subtract two matrices, they must have the SAME dimension!

$$\mathbf{A} + \mathbf{B} = [a_{ij}]_{m\times n} + [b_{ij}]_{m\times n} = [a_{ij} + b_{ij}]_{m\times n} \quad (2.7)$$

$$\mathbf{A} - \mathbf{B} = [a_{ij}]_{m\times n} - [b_{ij}]_{m\times n} = [a_{ij} - b_{ij}]_{m\times n} \quad (2.8)$$

This extends to more than two matrices.

$\rightarrow$ To multiply by a scalar simply multiply each element of the matrix by that number,

$$\alpha\mathbf{A} = \alpha[a_{ij}] = [\alpha a_{ij}] \quad (2.9)$$

Apart from the requirement on the dimension of the matrix, addition and subtraction act just like real numbers. We must be more specific with matrix multiplication.

3.1.2 Matrix Multiplication

To multiply two matrices the first matrix must have the same number of columns as the second matrix has rows (the column dimension of $\mathbf{A}$ must be equal to the row dimension of $\mathbf{B}$). If this is the case the matrices are said to be **conformable**. To multiply, take the first element of the first row of the first matrix and multiply by the first element of the first column of the second matrix, then the second element of the first row times the second element of the first column. Keep doing until you get to the end of the first row and first column and this is the first **entry** of the first row of the output matrix. Next you take the first row of the first matrix and the second column of the second matrix. Multiply the former by the latter, following what was explained above, and it gives the second element of the first row. After finishing the first row of the first matrix go to the second row of the first matrix and start again.

For example: Let $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$ In terms of dimension, $(2 \times 3) * (3 \times 2) = (2 \times 2)$. Obviously,

$$\begin{aligned} \mathbf{AB} &\neq \mathbf{BA} \\ (3 \times 2) * (2 \times 3) &= (3 \times 3) \end{aligned}$$

Check it with the two above matrices $\mathbf{A}$ and $\mathbf{B}$ (compute $\mathbf{BA}$).

In general, if $\mathbf{A}$ is of dimension $m \times n$ and $\mathbf{B}$ is of dimension $p \times q$, the matrix product $\mathbf{AB}$ will be defined if and only if $n = p$. If defined, moreover, the product matrix $\mathbf{AB}$ will have the dimension $m \times q$. So unlike numbers the **order of multiplication matters**. It matters if we **pre-multiply** or **post-multiply**. There are some rules for multiplying matrices but the most important thing to remember is that the order of multiplication matters.

$$\begin{aligned} (\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}) &= \mathbf{ABC} \\ \mathbf{A}(\mathbf{B} + \mathbf{C}) &= \mathbf{AB} + \mathbf{AC} \\ (\mathbf{A} + \mathbf{B})\mathbf{C} &= \mathbf{AC} + \mathbf{BC} \\ \mathbf{A}^n &= \mathbf{A} \cdot \mathbf{A} \cdots \mathbf{A} \end{aligned}$$

3.2 Some Special Matrices

3.2.1 The Identity Matrix

This serves as the number 1 in linear algebra. The identity matrix of order n is a square matrix with 1's on the *main diagonal and zeros everywhere else.*

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \ddots & 1 & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$$\mathbf{AI} = \mathbf{IA} = \mathbf{A}$$

Some properties: $\mathbf{AIB} = (\mathbf{AI})\mathbf{B} = \mathbf{AB}$
Note that $\mathbf{A}$ and $\mathbf{B}$ must still be conformable! It is also the case that

$$(\mathbf{I}_n)^k = I_n \text{ for } k = 1, 2, 3, \dots$$

3.2.2 The Null Matrix

A null matrix is a matrix whose elements are all zero. The zero matrix is not restricted to being square. It is denoted by $\mathbf{0}$.

Some properties:

$$\begin{aligned} \mathbf{A} + \mathbf{0} &= \mathbf{0} + \mathbf{A} = \mathbf{A} \\ \mathbf{A0} = \mathbf{0} &\text{ and } \mathbf{0A} = \mathbf{0} \end{aligned}$$

3.2.3 Idiosyncrasies of Matrix Algebra

For the case of scalars the equation $ab = 0$ always implies that either a or b is zero. ***This is not the case in matrix multiplication***. For instance, we have

$$\mathbf{AB} = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & 4 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \mathbf{0}$$

For the case of scalars the equation $cd = ce$ (with $c \neq 0$) always implies that $d = e$. ***This is not the case for matrices***. For instance, we have

$$\mathbf{C} = \begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \quad \mathbf{E} \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix}$$

We find that

$$\mathbf{CD} = \mathbf{CE} = \begin{bmatrix} 5 & 8 \\ 15 & 24 \end{bmatrix}$$

Even though $D \neq E$.

3.2.4 The Transpose

This is the operation of swapping the rows and the columns.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \Leftrightarrow \mathbf{A}' = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \ddots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{bmatrix}$$

Sometimes this is denoted as,

$$\mathbf{A}^T$$

Some properties

$$\begin{aligned} (\mathbf{A}')' &= \mathbf{A} \\ (\mathbf{A} + \mathbf{B})' &= \mathbf{A}' + \mathbf{B}' \\ (\mathbf{AB})' &= \mathbf{B}'\mathbf{A}' \end{aligned}$$

3.2.5 Symmetric Matrices

A symmetric matrix has the property that,

$$\mathbf{A} = \mathbf{A}'$$

That is, it is symmetric about the main diagonal.

3.3 The Inverse and the solution to a system of equations

If we can find a matrix such that,

$$\mathbf{AX} = \mathbf{XA} = \mathbf{I}$$

Then $\mathbf{X}$ is said to be the ***Inverse of*** $\mathbf{A}$ and is usually denoted by, $\mathbf{A}^{-1}$.

Some important points:

1. Not every square matrix has an inverse, squareness is a *necessary* condition, but not a *sufficient* condition for the existence of an inverse. If a square matrix $\mathbf{A}$ has an inverse, $\mathbf{A}$ is said to be *non-singular*; if $\mathbf{A}$ possesses no inverse, it is called a *singular* matrix.
2. If $\mathbf{A}^{-1}$ is invertible, and $(\mathbf{A}^{-1})^{-1} = A$.
3. If $\mathbf{A}$ is $n \times n$, then $\mathbf{A}^{-1}$ is $n \times n$.
4. If an inverse exists, then it is unique.
5. Let $\mathbf{A}$ and $\mathbf{B}$ be invertible $n \times n$ matrices. Then:
   (a) $\mathbf{AB}$ is invertible, and $(\mathbf{AB})^{-1} = B^{-1}A^{-1}$.
   (b) The transpose $\mathbf{A}'$ is invertible, and $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$.
   (c) $(c\mathbf{A})^{-}1 = c^{-1}\mathbf{A}^{-1}$ whenever c is a number $\neq 0$.

The Inverse is central to the solution of the system of equations. To find a solution one has to write the $\mathbf{x}$ variables in terms of the parameters only. Recall that our system is,

$$\mathbf{Ax} = \mathbf{b}$$

Then to find the solution we simply ***pre-multiply*** either side by $\mathbf{A}^{-1}$

$$\begin{aligned} \mathbf{A}^{-1}(\mathbf{Ax}) &= \mathbf{A}^{-1}\mathbf{b} \\ \mathbf{A}^{-1}\mathbf{Ax} &= \mathbf{A}^{-1}\mathbf{b} \\ \mathbf{Ix} &= \mathbf{A}^{-1}\mathbf{b} \\ \mathbf{x} &= \mathbf{A}^{-1}\mathbf{b} \end{aligned}$$

So ***all*** we need to do is find $\mathbf{A}^{-1}$ and pre-multiply the column-vector $\mathbf{b}$ by it to solve our system. Let's find the inverse of a general 2×2 matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Call this inverse $\mathbf{X}$

such that $\mathbf{AX} = \mathbf{I}$.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x & y \\ z & w \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Matrix multiplication implies that

$$ax + bz = 1, \qquad ay + bw = 0$$
$$cx + dz = 0, \qquad cy + dw = 1$$

Four equations and four unknowns:

$$x = \frac{d}{ad - bc},$$
$$z = \frac{-c}{ad - bc},$$
$$y = \frac{-b}{ad - bc},$$
$$w = \frac{a}{ad - bc}$$

Thus

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \Rightarrow \mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{2.10}$$

3.4 The Inverse and the Determinant

- The inverse might not exist, how can we tell?
- If the inverse exists how do we find it?

3.4.1 The Inverse, Non-Singularity and a Unique Solution

A ***square*** matrix whose ***rank*** equals the number of rows (or columns) is said to be ***non-singular***. If the ***coefficient matrix*** $\mathbf{A}$ is non-singular then $\mathbf{A}^{-1}$ exists and the system of linear equations has a unique solution.

- The general formula for the Inverse is,

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} adj(\mathbf{A})$$

Which exists (makes sense) iff,

$$|\mathbf{A}| \neq 0$$

The expression $|\mathbf{A}|$ is called the ***determinant*** and $adj(\mathbf{A})$ is the **adjoint matrix** which is a matrix of co-factors. Since the inverse only makes sense if the determinant exists it is perhaps not surprising that the system of linear equations only has a unique solution if the ***determinant exists***.

To calculate the determinant of a 2×2 matrix, subtract the off diagonal from the main diagonal.

$$\begin{vmatrix} 2 & 3 \\ \frac{1}{2} & 2 \end{vmatrix} = 4 - 1.5$$

The ***Minor*** of a $n \times n$ matrix $\mathbf{A}$ is the ***determinant of*** a $(n-1) \times (n-1)$ sub-matrix that is obtained by deleting a specific row and specific column of $\mathbf{A}$.

$$\mathbf{M}_{ij} = |\mathbf{A}_{ij}|$$

The **Co-factor** is then equal to,

$$\mathbf{C}_{ij} = (-1)^{i+j}\mathbf{M}_{ij}$$

The determinant of $n \times n$ matrix $\mathbf{A}$ is equal to,

$$|\mathbf{A}| = a_{11}\mathbf{C}_{11} + a_{12}\mathbf{C}_{12} + \cdots + a_{1n}\mathbf{C}_{1n} = a_{11}\mathbf{M}_{11} - a_{12}\mathbf{M}_{12} + \dots$$

Example:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} 2 & 1 & 0 \\ 1 & 5 & 0 \\ 1 & 0 & 2 \end{bmatrix} \\ |\mathbf{A}| &= 2\begin{vmatrix} 5 & 0 \\ 0 & 2 \end{vmatrix} - 1\begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} + 0\begin{vmatrix} 1 & 5 \\ 1 & 0 \end{vmatrix} \end{aligned}$$

We do not have to expand about the first row, any row or column will do. There are other simplifications too.

Rules for determinants of matrix A

1. If each of the entries in one or more rows or columns of $\mathbf{A}$ is 0 then $|\mathbf{A}| = 0$.

2. $|\mathbf{A}| = |\mathbf{A}'|$

3. If $\mathbf{B}$ is the matrix obtained by multiplying each entry of one row or column of $\mathbf{A}$ by the same number α, then $|\mathbf{B}| = \alpha|\mathbf{A}|$

4. If two rows or columns of $\mathbf{A}$ are equal $\Rightarrow |\mathbf{A}| = 0$

5. If two rows or columns of $\mathbf{A}$ are proportional $\Rightarrow |\mathbf{A}| = 0$

6. $|\mathbf{AB}| = |\mathbf{A}| \times |\mathbf{B}|$

7. If $\mathbf{A}_{n\times N}$ and $\alpha \in \mathbb{R} \Rightarrow |\alpha\mathbf{A}| = \alpha^n|\mathbf{A}|$

Finally for small systems of linear equations one can find the solution using ***Cramer's Rule***. ***Cramer's Rule*** states that the solution to the square system,

$$\begin{aligned} \mathbf{Ax} &= \mathbf{b} \\ x_i &= \frac{|\mathbf{D}_i|}{|\mathbf{A}|} \quad \forall i \end{aligned}$$

Where $\mathbf{D}_i$ is the matrix obtained by replacing the i^{th} column of $\mathbf{A}$ with the column vector $\mathbf{b}$.

3.5 Linear Independence

A system of linear equations has a unique solution ***iff*** the number of different equations equals the number of unknowns.

If it has more unknowns than equations it is impossible to write each variable as a function of its own equation that is free of other variables; there are simply not enough ***different equations***. On the other hand if you have more equations than unknowns and these equations are ***different***, then a solution will entail expressing at least one variable in terms of more than one equation, each one having ***different*** parameter values. This is impossible and the system has no solutions and is said to be ***inconsistent***.

I emphasize ***different*** because it is possible that a three-equation system in two variables is really a two-equation system in two variables. This occurs if one of the equations is a ***linear combination*** of the other two. Unfortunately this applies to a system that has the same number of equations as unknowns also. In this case the system really has more variables than equations and has no solution. At least one equation is said to be ***linearly dependent*** on some other equation. The ***rank of a matrix*** tells us how many rows of $\mathbf{A}$ are ***linearly***

independent. We would like it to be of full rank. This is what the determinant really tests. In the process of solving a square system of linear equations using matrix algebra, if the equations are linearly dependent one of the rows effectively contains only zeros so that the determinant must be zero.

When the squareness condition is already met, a sufficient condition for the nonsingularity of a matrix is that its rows be linearly independent (or equivalently its columns have to be linearly independent). Formally, an $n \times n$ coefficient matrix $\mathbf{A}$ can be considered as an ordered set of row vectors:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} v_1' \\ v_2' \\ \vdots \\ v_n' \end{bmatrix}$$

where $v_i' = [a_{i1} \ a_{i2} \ \cdots \ a_{in}]$, $i = 1, 2, \dots, n$. For the rows to be linearly independent, none must be a linear combination of the rest. Linear row independence requires that the only set of scalars c_i which can satisfy the vector equation

$$\sum_{i=1}^{n} c_i v_i' = \mathbf{0}$$

be $c_i = 0$ for all i.

→ If the maximum number of linearly independent rows that can be found in a matrix is r, the matrix is said to be of *rank* r.

Example 1: Prove that the vectors $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ are linearly independent.

→ In order to prove that let us write the following system of equations

$$c_1 + 2c_2 = 0 \tag{2.11}$$

$$c_2 + c_3 = 0 \tag{2.12}$$

$$c_1 + c_3 = 0 \tag{2.13}$$

From equations (2.11) and (2.12) we obtain that $c_1 = -c_3 = c_2$. Using the first equation this leads to $c_1 = 0 \Rightarrow c_2 = c_3 = 0$. We can conclude that the three vectors are linearly independent.

Theorem 5. *A set of n vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ in $\mathbb{R}^n$ is linearly independent if and only if*

$$det(\mathbf{v}_1 \; \mathbf{v}_2 \; \cdots \; \mathbf{v}_n) \neq 0.$$

Theorem 6. *If $k > n$, any set of k vectors in $\mathbb{R}^n$ is linearly dependent.*

3.6 An Econometric Application: Ordinary Least Squares in Matrix Form

One of the most important applications of linear algebra in economics is econometrics. In econometrics, we often want to estimate the relationship between one variable of interest and a set of explanatory variables. For example, suppose we want to understand how wages are related to education and experience. A simple linear model might be

$$w_i = \beta_0 + \beta_1 educ_i + \beta_2 exper_i + u_i,$$

where w_i is the wage of individual i, $educ_i$ is years of education, $exper_i$ is years of labor market experience, and u_i is an error term capturing factors not included in the model.

If we have data on n individuals, we can write the model as a system of n equations:

$$\begin{aligned} w_1 &= \beta_0 + \beta_1 educ_1 + \beta_2 exper_1 + u_1 \\ w_2 &= \beta_0 + \beta_1 educ_2 + \beta_2 exper_2 + u_2 \\ &\vdots \\ w_n &= \beta_0 + \beta_1 educ_n + \beta_2 exper_n + u_n. \end{aligned}$$

This system can be written compactly in matrix form as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where

$$\mathbf{y} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} 1 & educ_1 & exper_1 \\ 1 & educ_2 & exper_2 \\ \vdots & \vdots & \vdots \\ 1 & educ_n & exper_n \end{bmatrix}, \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}.$$

The first column of $\mathbf{X}$ consists of ones because the model includes a constant term β_0.

The Least Squares Problem

In practice, the points in the data will usually not lie exactly on a line or plane. Therefore, we cannot usually find a vector $\boldsymbol{\beta}$ that makes

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$$

hold exactly. Instead, we choose $\boldsymbol{\beta}$ to make the prediction errors as small as possible.

For any candidate coefficient vector $\boldsymbol{\beta}$, the fitted values are

$$\widehat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta},$$

and the residual vector is

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}.$$

Ordinary Least Squares chooses $\boldsymbol{\beta}$ to minimize the sum of squared residuals:

$$\min_{\boldsymbol{\beta}} \ \mathbf{e}'\mathbf{e} = \min_{\boldsymbol{\beta}} \ (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

The solution to this minimization problem is

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y},$$

provided that the inverse $(\mathbf{X}'\mathbf{X})^{-1}$ exists.

Why Linear Algebra Matters

The formula

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

uses several ideas from linear algebra:

- $\mathbf{X}'$ is the transpose of the data matrix $\mathbf{X}$.
- $\mathbf{X}'\mathbf{X}$ is a square matrix.
- The inverse $(\mathbf{X}'\mathbf{X})^{-1}$ exists only if $\mathbf{X}'\mathbf{X}$ is nonsingular.
- Nonsingularity requires that the columns of $\mathbf{X}$ be linearly independent.

The last point is especially important. If one explanatory variable is a linear combination of the others, then the columns of $\mathbf{X}$ are linearly dependent. In that case, $\mathbf{X}'\mathbf{X}$ is singular, the inverse does not exist, and the OLS formula is not well-defined.

Economic Interpretation of Linear Independence

Suppose we estimate the model

$$w_i = \beta_0 + \beta_1 educ_i + \beta_2 exper_i + u_i.$$

For the effects of education and experience to be separately identified, education and experience must provide different information in the data. If experience were always exactly twice education, so that

$$exper_i = 2educ_i$$

for every individual i, then the experience column of $\mathbf{X}$ would be a multiple of the education column. The two columns would be linearly dependent.

In that case, the data would not allow us to separately estimate the effect of education and the effect of experience. We could not tell whether wages are higher because of education, experience, or some exact linear combination of the two. This is why linear independence is not just a mathematical condition; it is also an economic and statistical identification condition.

A Small Example

Suppose we have data on three individuals and estimate

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

Let

$$\mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}.$$

Then

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix},$$

and

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 11 \\ 25 \end{bmatrix}.$$

Therefore,

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}^{-1} \begin{bmatrix} 11 \\ 25 \end{bmatrix}.$$

Since

$$\begin{vmatrix} 3 & 6 \\ 6 & 14 \end{vmatrix} = 3(14) - 6(6) = 42 - 36 = 6 \neq 0,$$

the matrix $\mathbf{X}'\mathbf{X}$ is nonsingular, so the inverse exists.

Using the formula for the inverse of a 2×2 matrix,

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}.$$

Thus,

$$\widehat{\boldsymbol{\beta}} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 11 \\ 25 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 3/2 \end{bmatrix}.$$

So the estimated regression line is

$$\widehat{y} = \frac{2}{3} + \frac{3}{2}x.$$

Main Takeaway

Linear algebra allows us to write many equations compactly, solve systems efficiently, and understand when econometric coefficients are well-defined. In particular, the rank and linear independence of the columns of $\mathbf{X}$ determine whether the OLS estimator can be computed uniquely.

4 Single Variable differentiation (on 07/15)

For reference, here are some common differentiation rules (these are in the front cover of a calculus book that I have had for 22 years and still reference from time to time – the meaning of this is to hold onto good books). Note that u and v are functions of x, and that c and n are constants.

4.1 A Brief Review of Univariate Calculus

When studying the graph of a function, we may want to have a measure of the steepness of a graph at one point. This is defined as the slope of the function at the particular point. Whereas this is easy for a line, it is more difficult if the graph is not a line.

Table 1: Derivatives

1. $\frac{d}{dx}[cu(x)] = cu'(x)$	2. $\frac{d}{dx}[u(x) \pm v(x)] = u'(x) \pm v'(x)$
3. $\frac{d}{dx}[u(x)v(x)] = u(x)v'(x) + v(x)u'(x)$	4. $\frac{d}{dx}\left[\frac{u(x)}{v(x)}\right] = \frac{v(x)u'(x)-u(x)v'(x)}{v(x)^2}$
5. $\frac{d}{dx}[c] = 0$	6. $\frac{d}{dx}[u(x)^n] = nu(x)^{n-1}u'(x)$
7. $\frac{d}{dx}[x] = 1$	8. $\frac{d}{dx}[\lvert u(x)\rvert] = \frac{u(x)}{\lvert u(x)\rvert}(u'(x)),\ u(x) \neq 0$
9. $\frac{d}{dx}[\ln u(x)] = \frac{u'(x)}{u(x)}$	10. $\frac{d}{dx}[e^{u(x)}] = e^{u(x)}u'(x)$
11. $\frac{d}{dx}[u[v(x)]] = u'[v(x)]v'(x)$ "The Chain Rule"	

4.1.1 The Slope of a linear Function

Let us define the function $y = f(x) = b + mx$. Consider two arbitrary distinct points, (x_0, y_0) and (x_1, y_1), belonging to the function f. The two arbitrary points must satisfy the function and therefore we have that $y_0 = b + mx_0$ and $y_1 = b + mx_1$. Then we obtain that $y_1 - y_0 = m(x_1 - x_0)$ and the slope is

$$\frac{\Delta y}{\Delta x} = \frac{y_1 - y_0}{x_1 - x_0} = m$$

The slope or m measures the rate of change of the function f.

4.1.2 The Slope of a Non-linear Function

Calculus is the ***approximation*** of a non-linear function using a ***function*** that is ***tangent*** (has the same slope) to any ***specific point*** of approximation, provided that function is ***continuous***.
As Δx gets smaller the approximation gets closer to the actual slope at the point of approximation x_0. The slope of a line at (x_0, y_0) is

$$\frac{\Delta y}{\Delta x} = \frac{y - y_0}{x - x_0} = \frac{f(x_0 + \Delta x) - f(x_0)}{(x_0 + \Delta x) - x_0}$$

The line that is just tangent to x_0 has exactly the same slope as the non-linear function at

x_0 and is equal to,

$$\frac{dy}{dx} = f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \left(\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \right) \tag{3.1}$$

This is the **derivative of y at x**. Notice that the derivative is contingent on the point of approximation. Differentiation yields a function that when evaluated at a specific point yields the slope of the tangent to the function at that point. The neat part is that the derivative can be evaluated at every point in the domain of the function for which it exists.

- The derivative (limit) in (3.3) may not exist. The requirement for a function to be differentiable is that it is ***continuous over the relevant interval***. Most functions used in economics are ***continuous*** and we will merely assume continuity. Roughly speaking continuity requires there are no breaks in the function or equivalently nearby points on the function are mapped into nearby points.

- If a function is differentiable, its derivative $f'(x)$ is also a function of x. This may also be differentiable and so on. A function that can be differentiated k times is said to be C^k, meaning continuously differentiable k times.

- There are several rules for differentiating single variable functions. It is important to know these. I recommend learning the chain rule well and the exponential and log rules.

4.1.3 The Second Derivative

The first derivative tells us how the dependent variable, $f(x)$ changes with x. The first derivative $f'(x)$ may itself be a function of x. It can be differentiated to see how it changes as x changes. Thus,

$$f''(x) \text{ is the change in } f'(x)$$

This second derivative contains a lot of important information like whether we have found a maximum, minimum, or inflection point.

4.2 First-derivative test for a relative extremum

If the first derivative of a function $f(x)$ at $x = x_0$ is $f'(x_0) = 0$, then the value of the function at x_0, $f(x_0)$, will be

a. A relative *maximum* if the derivative $f'(x)$ changes its sign from positive to negative from the immediate left of the point x_0 to its immediate right.

b. A relative *minimum* if the derivative $f'(x)$ changes its sign from negative to positive from the immediate left of the point x_0 to its immediate right.

c. Neither a relative maximum nor a relative minimum if $f'(x)$ has the same sign on both the immediate left and right of point x_0.

4.3 Second Order Sufficient Conditions

The first-derivative test is cumbersome. We can get can get the same information by looking at the second derivative. In analytical work we use the second derivative test evaluated at the *same point* as the first derivative. Because it is evaluated at the same point this is a test for *local extremum.*

The second derivative $f''(x_0)$ is the change in the slope $f'(x_0)$. This suggests that for a ***strict local maximum*** we require,

$$f'(x^*) = 0 \text{ and } f''(x^*) < 0 \tag{3.2}$$

Whilst for a ***strict local minimum***,

$$f'(x^*) = 0 \text{ and } f''(x^*) > 0 \tag{3.3}$$

If $f''(x^*) = 0$, the function is neither increasing nor decreasing at x^* which implies it isn't flat just at x^*. This doesn't mean it isn't a maximum, it means **we are not sure**.

To reiterate *the derivative test is sufficient only for a* ***local*** *maximum/minimum.*

For a global maximum/minimum we need to know more about the shape of the functions and this is where concavity and convexity of functions are important properties. However this is translated into the following conditions:

- c is a maximum point of f in I iff

$$f''(x) \leq 0 \text{ for all } x \in I, \text{ and } f'(c) = 0.$$

- c is a minimum point of f in I iff

$$f''(x) \geq 0 \text{ for all } x \in I, \text{ and } f'(c) = 0.$$

The second derivative plays a role in the curvature of the function. For the case of a maximum we need to check whether the function is **concave** whereas for a minimum we need to check whether the function is **convex**.

Examples:
1. Find the extreme point(s) of $f(x) = (x^2 - 2x)$ and determine if this is a local maximum or minimum. The above conditions yield

$$\begin{aligned} f'(x) &= 2x - 2 \\ f'(x^*) = 2x^* - 2 = 0 &\Rightarrow x^* = 1. \end{aligned}$$

The second derivative evaluated at our solution is then equal to $f''(x^*)$. Our solution is then a minimum. Note that for $-f(x) = -(x^2 - 2x)$ we find that $x^* = 1$ and $f''(x) = -2 < 0$, $x^* = 1$ is the maximum of $-f(x)$.

2. Find the extreme point(s) of $f(x) = x^3 - 3x$ and determine if this is a local maximum or minimum. The above conditions yield

$$\begin{aligned} f'(x) &= 3x^2 - 3 \\ f'(x^*) = 3(x^*)^2 - 3 = 0 &\Rightarrow x^* = \pm 1 \end{aligned}$$

The second derivative is then given by $f''(x) = 6x$. Therefore $x^* = 1$ is a minimum and $x = -1$ a maximum.

4.4 Implicit Differentiation

We know how to differentiate functions given by explicit formulas like $y = f(x)$. Now we consider how to differentiate functions defined implicitly by and equation such as $g(x, y) = c$, where c is a constant.

Simple case: Suppose $g(x, y) = xy = 5$. We say that $g(\cdot)$ defines y implicitly as a function of x. If we replace y with $f(x)$, then we can rewrite the function as

$$g(xy) \equiv xf(x) = 5$$

and then differentiate using the product rule:

$$1 \cdot f(x) + xf'(x) = 0$$

Thus solving for $f'(x)$ yields

$$f'(x) = -\frac{f(x)}{x}$$

or

$$y' = -\frac{y}{x}$$

In this particular case we can see that this is equivalent to the more direct way... from $g(\cdot)$ we know that this is equivalent to saying $y = \frac{5}{x}$, thus

$$\begin{aligned} y' &= -\frac{5}{x^2} \\ &= -\frac{5}{x} \end{aligned}$$

The Method of Implicit Differentiation
If two variables x and y are related by an equation, to find y':

1. Differentiate each side of the equation w.r.t. x, considering y as a function of x. (usually, you will need the chain rule.)

2. Solve the resulting equation for y'

Less Simple case: Suppose $y^3 + 3x^2y = 13$. Using implicit differentiation

$$\begin{aligned} 3y^2y' + 6xy + 3x^2y' = 0 \\ \therefore y' &= \frac{-6xy}{3x^2 + 3y^2} \\ &= \frac{-2xy}{x^2 + y^2} \end{aligned}$$

4.5 Approximations

4.5.1 The Differential

The derivative at x_0 defines the slope of the tangent at x_0. This tangent could be used to find a ***linear approximation*** to the relationship between y and x.

Given its slope (m), a line is defined as

$$y = b + mx$$

Thus we can approximate the tangent about x_0 by,

$$y = b + f'(x_0)x$$

This implies the change in y corresponding to an **assumed** change in x is,

$$\Delta y = f'(x_0)\Delta x$$

If Δx is big this isn't a very good approximation to the true change in y because the difference quotient approximation to the slope is only valid close to x_0. However for a small change, the approximation is better. It is called the ***differential*** and is represented by,

$$dy = f'(x_0)dx \tag{4.1}$$

Note also that (4.1) can be solved to give (3.1) and that if we assume $dx = 1$ then $dy = f'(x_0)$, the derivative is a good approximation to the change in y for a unit change in x.

Example: Derivative and Differential

$$y = x^3$$

Then:

$$\frac{dy}{dx} = 3x^2$$

This is a non-linear function of x but if $x = 2$,

$$\frac{dy}{dx} = 12$$

If $x = 3$,

$$\frac{dy}{dx} = 27$$

These are the slope of a tangent at the points $x = 2$ and $x = 3$ respectively.

The differential (linear approximation) at $x = 2$ is,

$$\Delta y = 12\Delta x$$

The second derivative is:

$$\frac{d^2y}{dx^2} = 6x$$

The third derivative is 6 and the fourth is zero. The function is C^4.

For derivative of higher order than 1 the following notation is usually used:
Second derivative: $f''(x) = \frac{d^2 f(x)}{dx^2}$ or $y'' = \frac{d^2 y}{dx^2}$

n^{th} derivative: $y^n(x) = f^n(x)$ or $\frac{d^n y}{dx^n}$

4.5.2 Higher order approximations

We have already talked about how the first derivative can be used to find an equation that is an linear approximation "around" a particular value a. Indeed for a linear equation, it is an exact representation... the approximation is given by

$$f(x) \approx f(a) + f'(a)(x - a)$$

Suppose $f(x) = b + mx$, then when $x = a$

$$\begin{aligned} f(x) &\approx b + ma + m(x - a) \\ &= b + mx \end{aligned}$$

this approximation is less good when $f(x)$ becomes more nonlinear. Thus mathematicians have come up with a **quadratic approximation**

Definition 13. *The* ***quadratic approximation*** *to $f(x)$ about $x = a$ is*

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2} f''(a)(x - a)^2 \text{ (x close to a)} \tag{4.2}$$

Suppose: $f(x) = x^{\frac{1}{3}}$ and we wanted to find the approximate value close to $x = 1$

$$\begin{aligned} f'(x) &= \frac{1}{3} x^{\frac{-2}{3}} \\ f''(x) &= \frac{-2}{9} x^{\frac{-5}{3}} \end{aligned}$$

Thus

$$f(x) \approx 1 + \frac{1}{3}(x - 1) - \frac{1}{9}(x - 1)^2$$

For example suppose $x = 1.03$

$$f(x) = 1.009901634049961 \qquad f(x) \approx 1 + \frac{1}{3}(1.03 - 1) - \frac{1}{9}(1.03 - 1)^2 = 1.0099$$

To get higher-order approximations:

Definition 14. *The **n-th order Taylor polynomial** for $f(x)$ about $x = a$ is*

$$f(x) \approx f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^n(a)}{n!}(x-a)^n \tag{4.3}$$

4.6 Properties of exponentials and LOG

1. $e^{\ln x} = x$ and $\ln e^x = x$
2. $a^r * a^s = a^{r+s}$
3. $a^{-r} = 1/a^r$
4. $a^r/a^s = a^{r-s}$
5. $(a^r)^s = a^{rs}$
6. $a^0 = 1$
7. $\log(r * s) = \log r + \log s$
8. $\log(1/s) = -\log s$
9. $\log(r/s) = \log r - \log s$
10. $\log(r^s) = s \log r$
11. $\log 1 = 0$

5 Concavity and Convexity (on 07/15 and 07/16)

The second order conditions restrictions are concerned with the curvature of the function. Concavity/Convexity is the name given to the curvature of functions. Knowing the curvature of functions is useful and not just for the second order conditions. For example knowing the shape of the expected utility function can tell us if an individual is risk averse or not. Concavity/Convexity of a function is related to but not the same thing as a convex set. We will concentrate on Concavity/Convexity of functions in terms of the second order conditions.

5.1 Definition of a Concave function

A concave function has a ***decreasing slope***. A function of one variable is ***concave*** if,

$$f''(x) < 0$$

A ***convex*** function has an increasing slope,

$$f''(x) > 0$$

The connection with the second order condition is obvious.

Another way of stating the above is to say that a function is ***concave (convex)*** if a secant line between any two points on the function is always ***below (above)*** the function.

5.2 The Line Definition of Concavity and Convexity in $\mathbb{R}^2$

We could write the x value between a and b as a function of the ***parameter*** t where, $0 \leq t \leq 1$,

$$x(t) = [1-t]a + tb$$

Then

$$f(x(t)) = f([1-t]a + tb)$$

As we vary the value of t between $0 \leq t \leq 1$ we vary the value of x between the two endpoints.

Next notice that we can write any (secant) line between two arbitrary points on the function as,

$$l = (1-t)f(a) + tf(b)$$

If, except at the endpoints the function is always above the line,

$$f([1-t]a + tb) \geq (1-t)f(a) + tf(b)$$

for any value of $t \in [0, 1]$ we know that the function is concave over the entire interval. To test for convexity reverse the inequality.

5.3 Concavity and Convexity in $\mathbb{R}^n$

In $\mathbb{R}^n$ we need two points on the function associated with two points in the *n-dimension* domain. Thus a secant line between the n dimensional points $\mathbf{a}$ and $\mathbf{b}$ on $y = f(\mathbf{x})$ can be written as,

$$l(\mathbf{a}, \mathbf{b}) = (1-t)f(\mathbf{a}) + tf(\mathbf{b}) \quad 0 \leq t \leq 1$$

Likewise the domain of $\mathbf{x}$ over $\mathbf{a}$, $\mathbf{b}$ is,

$$\mathbf{x} = (1-t)\mathbf{a} + t\mathbf{b} \quad 0 \leq t \leq 1$$

The function as,

$$f(\mathbf{x}) = f((1-t)\mathbf{a} + t\mathbf{b})$$

Definition A function is **Concave** on $(\mathbf{a}, \mathbf{b})$ if,

$$f([1-t]\mathbf{a} + t\mathbf{b}) \geq (1-t)f(\mathbf{a}) + tf(\mathbf{b}) \quad 0 \leq t \leq 1$$

A function is Convex on if,

$$f([1-t]\mathbf{a} + t\mathbf{b}) \leq (1-t)f(\mathbf{a}) + tf(\mathbf{b}) \quad 0 \leq t \leq 1$$

If these hold with ***strict*** inequalities except at the end points then the function is ***strictly concave or strictly convex***. (By strict we mean that inside $(\mathbf{a}, \mathbf{b})$ we can omit the equal signs sign).

5.4 Convex Sets

Don't confuse a convex set with a convex function. A convex set is some collection of objects for which,

• If $\mathbf{a}$ and $\mathbf{b}$ both belong to a set S then a line joining any two points in the set also belongs to the set S.

$$(1-t)\mathbf{a} + t\mathbf{b} \in S \quad 0 \leq t \leq 1$$

Notice this is just a ***weighted average*** of two points in a set.

The definition of a convex set requires that for any members of the set can be written as a linear (convex) combination of any other two members. This is just another way of saying weighted average.

Both concave and convex functions have convex sets as their domains. In $\mathbb{R}^2$ the domain is the relevant interval over the x axis which forms a convex set. More generally the definition of concavity and convexity require the function to be defined on,

$$f(\mathbf{x}) = f((1-t)\mathbf{a} + t\mathbf{b})$$

Therefore, they implicitly assume the domain of $\mathbf{x}$ is convex.

Useful properties on Convex and Concave functions

Let f and g be functions defined over a convex set S in $\mathbb{R}^n$. Then

1. f and g concave and $a \geq 0,\ b \geq 0 \Rightarrow af + bg$ is concave.
2. f and g convex and $a \geq 0,\ b \geq 0 \Rightarrow af + bg$ is convex.
3. $f(x)$ concave and $F(u)$ concave and increasing $\Rightarrow U(x) = F(f(x))$ is concave.
4. $f(x)$ convex and $F(u)$ convex and increasing $\Rightarrow U(x) = F(f(x))$ is convex.
5. f and g concave $\Rightarrow h(x) = \min\{f(x), g(x)\}$ is concave.
6. f and g convex $\Rightarrow h(x) = \max\{f(x), g(x)\}$ is convex.

5.5 Linear Approximation and Small Policy Changes

One of the most useful applications of derivatives in economics is approximation. Often, we know the value of a function at some point and want to know how the function changes when the input changes slightly. Instead of fully recomputing the function, we can use the derivative to construct a local linear approximation.

Let $y = f(x)$ be differentiable at x_0. For values of x close to x_0, we can approximate $f(x)$ by the equation of the tangent line at x_0:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

Equivalently, if $\Delta x = x - x_0$, then

$$\Delta y \approx f'(x_0)\Delta x.$$

Thus, the derivative $f'(x_0)$ tells us the approximate change in y caused by a small change in x.

Economic Interpretation

In economics, $f'(x_0)$ is often interpreted as a marginal effect. The approximation

$$\Delta y \approx f'(x_0)\Delta x$$

says that the total change in y is approximately equal to the marginal effect at the initial point multiplied by the size of the change in x.

This is especially useful when studying small policy changes, such as a small tax increase, a small price change, or a small change in income.

Example: Demand and a Small Price Change

Suppose demand for a good is given by

$$Q(p) = 100 - 4p^{1/2}.$$

Assume the current price is $p_0 = 25$. Then current demand is

$$Q(25) = 100 - 4(25)^{1/2} = 100 - 4(5) = 80.$$

The derivative of demand with respect to price is

$$Q'(p) = -2p^{-1/2}.$$

Evaluating this derivative at $p_0 = 25$ gives

$$Q'(25) = -2(25)^{-1/2} = -\frac{2}{5}.$$

Now suppose price rises from 25 to 26, so that

$$\Delta p = 1.$$

Using the linear approximation,

$$\Delta Q \approx Q'(25)\Delta p = -\frac{2}{5}(1) = -0.4.$$

Therefore,

$$Q(26) \approx Q(25) - 0.4 = 79.6.$$

So a small price increase from 25 to 26 reduces demand by approximately 0.4 units.

Exact Change versus Approximate Change

The exact change is

$$Q(26) - Q(25) = \left[100 - 4(26)^{1/2}\right] - 80.$$

The approximation avoids having to compute this exactly. For small changes in p, the approximation is usually close to the true value.

5.6 Comparative Statics and Linear Approximation

One of the central uses of derivatives in economics is *comparative statics*. Comparative statics asks how an economic outcome changes when some parameter of the model changes. For example, we may want to know how a firm's optimal output changes when a tax changes, how a consumer's demand changes when income changes, or how equilibrium price changes when demand shifts.

The basic idea is simple. Suppose an economic outcome y depends on some parameter a:

$$y = f(a).$$

If a changes slightly from a_0 to $a_0 + \Delta a$, then the exact change in y is

$$\Delta y = f(a_0 + \Delta a) - f(a_0).$$

If Δa is small, we can approximate this change using the derivative:

$$\Delta y \approx f'(a_0)\Delta a.$$

Equivalently,

$$f(a_0 + \Delta a) \approx f(a_0) + f'(a_0)\Delta a.$$

Thus, the derivative $f'(a_0)$ tells us the approximate effect of a small change in the parameter a on the economic outcome y.

Example 1: A Firm's Output and a Per-Unit Tax

Suppose a firm sells output at market price p and has cost function $C(q)$. The government imposes a per-unit tax t, so the firm's profit is

$$\pi(q, t) = pq - C(q) - tq.$$

The firm chooses q to maximize profit:

$$\max_q \ \pi(q, t).$$

The first-order condition is

$$\frac{\partial \pi(q, t)}{\partial q} = p - C'(q) - t = 0.$$

Let the optimal output be denoted by $q^*(t)$. Then $q^*(t)$ satisfies

$$p - C'(q^*(t)) - t = 0.$$

This equation implicitly defines optimal output as a function of the tax.

To find how optimal output changes when the tax changes, differentiate both sides with respect to t:

$$-C''(q^*(t))\frac{dq^*}{dt} - 1 = 0.$$

Solving for $\frac{dq^*}{dt}$ gives

$$\frac{dq^*}{dt} = -\frac{1}{C''(q^*(t))}.$$

If $C''(q) > 0$, then

$$\frac{dq^*}{dt} < 0.$$

Thus, an increase in the per-unit tax reduces the firm's optimal output.

Linear Approximation

Suppose the current tax is t_0. If the tax changes by a small amount Δt, then

$$q^*(t_0 + \Delta t) \approx q^*(t_0) + \frac{dq^*}{dt}\Delta t.$$

This approximation allows us to estimate the effect of a tax change without solving the firm's maximization problem again.

Numerical Example

Suppose

$$p = 100$$

and

$$C(q) = 20q + q^2.$$

Then

$$\pi(q,t) = 100q - (20q + q^2) - tq.$$

The first-order condition is

$$100 - 20 - 2q - t = 0.$$

Therefore,

$$80 - 2q - t = 0.$$

Solving for optimal output,

$$q^*(t) = 40 - \frac{t}{2}.$$

So

$$\frac{dq^*}{dt} = -\frac{1}{2}.$$

This means that a one-unit increase in the tax reduces optimal output by one-half unit.

Suppose the current tax is $t_0 = 10$. Then

$$q^*(10) = 40 - \frac{10}{2} = 35.$$

If the tax rises from 10 to 12, then $\Delta t = 2$. Using linear approximation,

$$\Delta q^* \approx \frac{dq^*}{dt}\Delta t = -\frac{1}{2}(2) = -1.$$

Therefore,

$$q^*(12) \approx 35 - 1 = 34.$$

In this example, the approximation is exact because $q^*(t)$ is linear in t.

Example 2: Demand and Income

Suppose demand for a good depends on income m:

$$x(m) = 5 + 2m^{1/2}.$$

The derivative is

$$x'(m) = m^{-1/2}.$$

If current income is $m_0 = 100$, then

$$x'(100) = \frac{1}{10}.$$

Suppose income increases by $\Delta m = 5$. Then the approximate change in demand is

$$\Delta x \approx x'(100)\Delta m = \frac{1}{10}(5) = 0.5.$$

Therefore,

$$x(105) \approx x(100) + 0.5.$$

Since

$$x(100) = 5 + 2(10) = 25,$$

we get

$$x(105) \approx 25.5.$$

Thus, a small increase in income from 100 to 105 increases demand by approximately 0.5 units.

Main Takeaway

Comparative statics studies how economic outcomes change when parameters change. Derivatives give the marginal comparative static effect. Linear approximation turns that marginal effect into an approximate total effect:

$$\Delta y \approx \frac{dy}{da}\Delta a.$$

This is useful because economists often want to predict the effect of small changes in taxes, prices, income, technology, or other parameters without solving the full model again. "'

Main Takeaway

Linear approximation says that a nonlinear function can be treated like a straight line when we stay close to a given point. In economics, this allows us to approximate the effect of small changes in prices, taxes, income, or inputs using marginal effects. "'

6 Integration (on 07/16 and 07/17)

For reference, here are some common integration rules. Note that C and k are constants. The integral is also referred to as the "anti-derivative" because if $A(x) = \int_0^x f(t)dt$ then $A'(x) = f(x)$. The most useful aspect of integrals is that they find the area under a line. This is particularly helpful when finding consumer surplus, producer surplus, and stochastic functions.

Table 2: Integrals

1. $\int kf(u)du = k\int f(u)du$	2. $\int [f(u) \pm g(u)]du = \int f(u)du \pm \int g(u)du$
3. $\int du = u + C$	4. $\int u^n du = \frac{u^{n+1}}{n+1} + C,\ n \neq -1$
5. $\int \frac{du}{u} = \ln\|u\| + C$	6. $\int e^u du = e^u + C$

6.1 Fundamental Theorems of Calculus

Theorem 7. *If a function f is continuous on the closed interval $[a, b]$, then*

$$\int_a^b f(x)dx = F(b) - F(a)$$

where F is any function that $F'(x) = f(x)$ for all $x \in [a, b]$.

Theorem 8. *If f is continuous and $a(x)$ and $b(x)$ are differentiable, then*

$$\frac{d}{dx}\left[\int_{a(x)}^{b(x)} f(t)dt\right] = f(b(x))b'(x) - f(a(x))a'(x).$$

6.2 Example 1: Consumer Surplus

Suppose we have the following inverse demand function:

$$p(q) = 100 - 2q$$

and we know that the equilibrium quantity bought is $Q = 30$ and corresponding price is $p(Q) = 40$. Thus consumer surplus is the difference between what a consumer is willing to pay and what she actually pays.

$$
\begin{aligned}
CS &= \int_0^Q p(q) - p(Q)dq \\
&= \int_0^Q p(q)dq - \int_0^Q p(Q)dq \\
&= \int_0^Q p(q)dq - p(Q)Q \\
&= \int_0^{30} 100 - 2qdq - (30)(40) \\
&= 100(30) - (30)^2 - (30)(40) \\
&= 30(100 - 30 - 40) \\
&= 30(30) = \frac{1}{2}(30)(60) = \frac{1}{2} \times b \times h \\
&= \$900
\end{aligned}
$$

6.2.1 Total Welfare

What if we wanted to find total social welfare, that is Consumer Surplus plus Producer Surplus? First suppose the inverse supply function is

$$\hat{p}(q) = 10 + q$$

Now we need to find the new equilibrium point. Setting $\hat{p}(q) = p(q)$ we find that the equilibrium quantity is $Q = 30$ (the same as before – through dumb luck actually). Thus the total social welfare is

$$
\begin{aligned}
CS + PS &= \int_0^Q p(q) - p(Q)dq + \int_0^Q p(Q) - \hat{p}(q)dq \\
&= \int_0^Q p(q)dq - p(Q)Q + p(Q)Q - \int_0^Q \hat{p}(q)dq \\
&= \int_0^Q p(q)dq - \int_0^Q \hat{p}(q)dq \\
&= \int_0^{30} 100 - 2qdq - \int_0^{30} 10 + qdq \\
&= 100(30) - (30)^2 - 10(30) - \frac{1}{2}(30)^2 \\
&= 30(45) = \frac{1}{2}(30)(90) = \frac{1}{2} \times b \times h \\
&= \$1,350
\end{aligned}
$$

6.3 Random Variables and Expectations

Let X = random variable with probability density function (pdf) $f(x),\ l \leq x \leq u$.[5] Then the $Pr(x \leq X \leq x + dx) = f(x)dx$. However dx is quite small. If you want do find the probability the random variable lies in between some interval $[a, b]$, then you would need to integrate

$$Pr(a \leq X \leq b) = \int_a^b f(x)dx$$

The expected value (sometimes referred to as the mean or μ) is

$$E[X] = \int_l^u xf(x)dx.$$

The expectation of a function of X (think expected utility) is

$$E[u(X)] = \int_l^u u(x)f(x)dX$$

Important Side note: Expected utility is NOT the utility of the expected value!! It is almost always the case that

$$E[u(X)] \neq u(E(X))$$

The Cumulative Distribution Function (CDF) is

$$F(x) = Pr(X \leq x) = \int_l^x f(t)dt$$

7 Multivariate Calculus (on 07/17)

To represent a function of one variable we need two dimensions (or the plane); a function of two variables needs three dimensions and so on. However we are limited geometrically to three dimensions and so most of the principles of multivariate calculus are illustrated using two variables. In $\mathbb{R}^3$ the value of $z = f(x, y)$ is the height of the function above the (x, y) plane. Because the function is defined for each x over all y values (in the domain) and vice versa the graph is a ***surface***. In higher dimensions the graphs of functions are referred to as surfaces or curves. We know how to do ***level curves***. The idea is simple, consider

$$z = f(x, y)$$

[5]Here, l represents the lower bound and u the upper bound. In most cases these will be equal to $-\infty$ and ∞ respectively.

If we fix the value of z this becomes a one variable function (we can choose one of the variables then the other must ensure the right hand side equals z) and can be graphed in the plane. If we do this for different values of z we in effect get the graph of $z = f(x, y)$ in two dimensions. For example the utility function $U = F(x_1, x_2)$ has 2 independent variables and can be graphed using indifference curves. An ***indifference curve*** tells us the combinations of x_1 and x_2 that result in the same amount of utility.

Example: Consider the utility function

$$U = x_1^{\frac{1}{2}} x_2^{\frac{1}{3}}$$

The level curve for $U = 1$ is $x_2 = x_1^{\frac{-3}{2}}$, and for $U = 2$ it is $x_2 = 8x_1^{\frac{-3}{2}}$.

7.1 A preview of the main points

In $\mathbb{R}^3$, a function with a unique maximum looks like the surface of a real 3-dimensional hill. Suppose we wish to know where the hill is at its maximum. Evidently, at the top, the hill is flat and so the derivative is zero in ***every*** direction. We could differentiate z by x holding *y constant*, or we could differentiate z by y holding *x constant*. The former moves us in a direction parallel to the x-axis and is called the partial derivative with respect to x and the latter moves us in a direction parallel to the y-axis and is called the partial derivative with respect to y. This is good because if we are at the top of the hill both of these will have to be zero. This is *necessary* but not *sufficient*. In particular the bottom of a bowl would also be flat as would any 'dips' in a function.

When we compute the partial derivative of z with respect to x, we measure the rate of change of the function as x changes and y remains constant. Our second order conditions involve the second order derivatives. Finally, notice we are moving only in directions parallel to the x, y axes. On our real hill we can move in all directions (we may off course fall off) but if we can define a line in $\mathbb{R}^n$ we can also differentiate along this line, this is known as the *directional derivative*.

7.2 Partial Derivatives

Let $y = f(x_1, x_2, \ldots, x_n) = f(\mathbf{x})$ then,

$$\frac{\partial f(\cdot)}{\partial x_i}$$

is the **partial derivative of y with respect x_i** when all the other variables are held **constant**. This is formally defined as

$$\frac{\partial y}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, x_2, \ldots, x_i + h, \ldots, x_n) - f(x_1, x_2, \ldots, x_i, x_n)}{h}$$

When computing partial derivatives we must hold $(n-1)$ independent variables constant while allowing one variable to vary. We then apply the rules we learned when differentiating functions of one variable.

7.3 Vector Notation

In vector notation we would write,

$$\begin{aligned} y &= f(\mathbf{x}) \\ \mathbf{x} &= (x_1, x_2, \ldots, x_n) \end{aligned}$$

The partial derivatives could be written as,

$$\begin{aligned} f_{x_1}(\mathbf{x}) &= \frac{\partial f}{\partial x_1} \\ f_{x_1 x_1}(\mathbf{x}) &= \frac{\partial^2 f}{\partial x_1^2} \\ f_{x_1 x_2}(\mathbf{x}) &= \frac{\partial^2 f}{\partial x_1 \partial x_2} \end{aligned}$$

Every n differentiable variable function has n first order partials (may equal zero). Each of these first order partials is potentially a function of the n variables and so has n partials. So there are n^2 second order partials. These can be stacked into a ***Hessian Matrix***. For example the Hessian matrix for a function of 3 variables is.

$$\mathbf{H} = \begin{bmatrix} f_{x_1 x_1} & f_{x_1 x_2} & f_{x_1 x_3} \\ f_{x_2 x_1} & f_{x_2 x_2} & f_{x_2 x_3} \\ f_{x_3 x_1} & f_{x_3 x_2} & f_{x_3 x_3} \end{bmatrix} = \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \tag{7.1}$$

By ***Young's theorem*** the cross derivative terms are equal to one another and so the Hessian matrix is symmetric. This matrix is important in checking conditions for maximization or minimization of multivariate functions. An implication of Young's theorem is that the order of differentiation does not matter when computing derivatives.

The row vector known as the ***Jacobian*** is,

$$\mathbf{J} = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} & \frac{\partial f(\mathbf{x})}{\partial x_2} & \dots & \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix} = \begin{bmatrix} f_1 & f_2 & \dots & f_n \end{bmatrix} \tag{7.2}$$

Example:

$$Y = AK^{\alpha}L^{1-\alpha}$$

Then:

$$\begin{aligned} \mathbf{J} &= \begin{bmatrix} Y_K & Y_L \end{bmatrix} = \begin{bmatrix} \alpha AK^{\alpha-1}L^{1-\alpha} & (1-\alpha)AK^{\alpha}L^{-\alpha} \end{bmatrix} \\ \mathbf{H} &= \begin{bmatrix} (\alpha-1)\alpha AK^{\alpha-2}L^{1-\alpha} & (1-\alpha)\alpha AK^{\alpha-1}L^{-\alpha} \\ (1-\alpha)\alpha AK^{\alpha-1}L^{-\alpha} & (\alpha-1)\alpha AK^{\alpha}L^{-(\alpha+1)} \end{bmatrix} \end{aligned}$$

Notice that this satisfies Young's theorem.

Suppose that $\alpha = \frac{1}{3},\ A = 4,\ K = 625,\ L = 10,000$. What happens if capital and labor are increased by 10 units each? To answer this exactly we need to calculate the following,

$$\Delta Y = F(K + \Delta K, L + \Delta L) - F(K, L) = 4(635)^{\frac{1}{3}}(10,010)^{\frac{2}{3}} - 4(625)^{\frac{1}{3}}(10,000)^{\frac{2}{3}} \approx 94.85$$

The differential approximation evaluated at $\alpha = \frac{1}{3},\ A = 4,\ K = 625,\ L = 10,000$ is,

$$\Delta Y = \frac{\partial Y(K,L)}{\partial K}\Delta K + \frac{\partial Y(K,L)}{\partial L}\Delta L = (1.5)10 + 810 = 95$$

The error is not too large but for bigger changes in the inputs the error grows.

7.4 The Chain Rule

A ***composite*** function is a function of one or more variables in which the variables are themselves functions of other variables. For example in a production function output will be a function of capital and labor which evolve over time.

$$Y = F(K(t), L(t))$$

The *Chain rule* tells us how to differentiate functions of this type in a very logical way.

$$\frac{dY}{dt} = \frac{\partial F(K,L)}{\partial K}\frac{dK}{dt} + \frac{\partial F(K,L)}{\partial L}\frac{dL}{dt} \tag{7.3}$$

This is referred to as the ***total derivative of Y with respect to t***. This extends naturally to n-variables.

7.5 The Implicit Function Theorem

Sometimes we may wish to differentiate a function that is not easy to solve explicitly in terms of the dependent variable. The functions can be defined implicitly by one equation or by a system of simultaneous equations. For the former case, we have, for instance,

$$F(x, y) = c$$

with c being a constant. Thus the previous equation defines a level curve for F. Suppose this defines y as an implicit function of x. We are interested in knowing the rate of change of y with respect to x, namely $\frac{dy}{dx}$. Let us rewrite our original equation replacing that y is an ***implicit function*** of x, then we get

$$F(x, f(x)) = c$$

Applying the chain rule, we obtain

$$\frac{\partial F}{\partial x}\frac{dx}{dx} + \frac{\partial F}{\partial f(x)}\frac{df(x)}{dx} = \frac{dc}{dx} = 0$$

If we solve this and use $y = f(x)$,

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y} \tag{7.4}$$

This is the ***implicit function theorem*** for one variable. If $\frac{\partial F}{\partial y} = 0$ this is not defined and thus the implicit function $y = f(x)$ does not exist. Economists often call the fraction $-\frac{dy}{dx}$ the marginal rate of substitution between y and x.

Example:

1. Compute $\frac{dy}{dx}$ given that $xy^{\frac{1}{2}} = 2$. Using the above we obtain

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y} = -\frac{y^{\frac{1}{2}}}{\frac{1}{2}xy^{-\frac{1}{2}}} = -\frac{2y}{x}$$

General case

For $z = F(x_1, x_2, \ldots, x_n)$ we have,

$$\frac{dx_j}{dx_i} = -\frac{\partial F/\partial x_i}{\partial F/\partial x_j}$$

This is defined if the denominator exists.

7.6 Elasticities

1. Elasticity measures how responsive one variable is to changes in another
2. The more elastic, the higher the responsiveness

The Price Elasticity of Demand is the percentage change in quantity demanded divided by the percentage change in price.

$$\epsilon_d = \frac{\frac{|Q_2 - Q_1|}{Q_1}}{\frac{|P_2 - P_1|}{P_1}} = \frac{|Q_2 - Q_1|}{Q_1} \times \frac{P_1}{|P_2 - P_1|} = \frac{|\Delta Q|}{|\Delta P|} \times \frac{P_1}{Q_1} = \frac{dQ}{dP}\left(\frac{P}{Q}\right)$$

$\rightarrow$ Elastic Demand is where $\epsilon_d > 1$. $\rightarrow$ Unit Elastic Demand is where $\epsilon_d = 1$. $\rightarrow$ Inelastic Demand is where $\epsilon_d < 1$. In general

$$\epsilon_{yx} = \frac{x}{y}\frac{dy}{dx} \tag{7.5}$$

We can easily translate this to the multi-variable case. Suppose we have a function of two variables $y(x, z)$

$$\epsilon_{yx} = \frac{x}{y}\frac{\partial y}{\partial x} \qquad \epsilon_{yz} = \frac{z}{y}\frac{\partial y}{\partial z}$$

Example Suppose we have a demand function

$$q(p, w) = \frac{\alpha w}{p} \tag{7.6}$$

Then the price elasticity of demand and income elasticity of demand is

$$\begin{aligned}
\epsilon_{qp} &= \frac{p}{q}\frac{\partial q}{\partial p} = \frac{p}{q}\left(\frac{-\alpha w}{p^2}\right) = \frac{p}{\frac{\alpha w}{p}}\left(\frac{-\alpha w}{p^2}\right) = -1 \\
\epsilon_{qw} &= \frac{w}{q}\frac{\partial q}{\partial w} = \frac{w}{\frac{\alpha w}{p}}\left(\frac{\alpha}{p}\right) = 1
\end{aligned}$$

8 Unconstrained Optimization (on 07/21)

This is what economics is all about...**Optimization**. We just talked about taking derivatives, or finding the slope of functions. Now, we can use this tool to find optimal values.

A function $f(\mathbf{x})$ has a ***global maximum*** at c if,

$$f(\mathbf{x}) \leq f(c)\ \forall x \in D$$

It has a ***global minimum*** at d if,

$$f(\mathbf{x}) \geq f(d)\ \forall x \in D$$

The value $f(c)$ is called the maximum value whereas $f(d)$ is called the minimum value. If the value of f at c is strictly larger than at any other point in D, then c is a strict maximum point. Similarly, d is a strict minimum point if $f(x) > f(d)$ for all $x \in D$, $x \neq d$.

If we restrict ourselves to analyzing points nearby to $\mathbf{x}^*$ (either c or d) then we talk of ***local maxima and minima***.

- Any minimization problem can easily be rewritten as a maximization problem since if $f(\mathbf{x}) \leq f(\mathbf{x}^*)$ then,
$$-f(\mathbf{x}) \geq -f(\mathbf{x}^*)$$

If $\mathbf{x}^*$ is the maximum of $f(\mathbf{x})$ it is also the minimum of $-f(\mathbf{x})$ therefore,

$$\min\{f(\mathbf{x})\} \text{ and } \max\{-f(\mathbf{x})\} \tag{10.1}$$

will yield the same $\mathbf{x}^*$.

8.1 The First Order Condition for all functions

Let $y = f(x_1, x_2, \ldots, x_n) = f(\mathbf{x})$. A **Necessary** but **NOT Sufficient** condition for a maximum or minimum is,

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = 0\ \forall i \tag{10.2}$$

$$Df(\mathbf{x}) = 0 \tag{10.3}$$

This is the ***first order condition*** and it will find a flat segment of any given function. The point verifying (10.2) or (10.3) is called a ***critical value***. Stationary points occur, for the single case variable, where the tangent to the graph of the function is parallel to the x-axis.

8.2 Second Order Sufficient Conditions

The first-derivative test is cumbersome. We can get can get the same information by looking at the second derivative. In analytical work we use the second derivative test evaluated at the *same point* as the first derivative. Because it is evaluated at the same point this is a test for *local extremum.*

The second derivative $f''(x_0)$ is the change in the slope $f'(x_0)$. This suggests that for a ***strict local maximum*** we require,

$$f'(x^*) = 0 \text{ and } f''(x^*) < 0 \tag{10.4}$$

Whilst for a ***strict local minimum***,

$$f'(x^*) = 0 \text{ and } f''(x^*) > 0 \tag{10.5}$$

If $f''(x^*) = 0$, the function is neither increasing nor decreasing at x^* which implies it isn't flat just at x^*. This doesn't mean it isn't a maximum, it means **we are not sure**.

To reiterate *the derivative test is sufficient only for a* ***local*** *maximum/minimum.*
Examples:
1. Find the extreme point(s) of $f(x) = (x^2 - 2x)$ and determine if this is a local maximum or minimum. The above conditions yield

$$\begin{aligned} f'(x) &= 2x - 2 \\ f'(x^*) = 2x^* - 2 = 0 &\Rightarrow x^* = 1. \end{aligned}$$

The second derivative evaluated at our solution is then equal to $f''(x^*)$. Our solution is then a minimum. Note that for $-f(x) = -(x^2 - 2x)$ we find that $x^* = 1$ and $f''(x) = -2 < 0$, $x^* = 1$ is the maximum of $-f(x)$.

2. Find the extreme point(s) of $f(x) = x^3 - 3x$ and determine if this is a local maximum or

minimum. The above conditions yield

$$\begin{aligned} f'(x) &= 3x^2 - 3 \\ f'(x^*) = 3(x^*)^2 - 3 = 0 &\Rightarrow x^* = \pm 1 \end{aligned}$$

The second derivative is then given by $f''(x) = 6x$. Therefore $x^* = 1$ is a minimum and $x = -1$ a maximum.

8.2.1 Second order Conditions for Multivariate Functions

The n-dimensional version of $f''(x) \leq 0$ ($f''(x) \geq 0$) is expressed in terms of the n-dimensional differential, that the Hessian (or bordered Hessian) matrix H, is negative semi-definite (P.S.D), but the latter is also exactly the condition for the objective function to be weakly concave (convex).

It is logical to suppose that the second order conditions involve second derivatives. The difficulty is, with n variables there are n^2 second derivatives. To put it another way we can move in various directions over the top of the hill, not just directions parallel to the (x, y) axes. For a maximum it is necessary that the first order partials in these directions are zero.

The S.O.C. for two variable functions

Reconsider the differential at some given point for a univariate function,

$$dy = f'(x)dx$$

Intuitively the test $f'(x) = 0$ is the same as $dy = 0$. The derivative $f'(x)$ is a function of x, but dx is an assumed change in the *choice variable* x, meaning it is free and we can treat it as a constant. Taking the second differential,

$$d^2y = \frac{\partial}{\partial x}\big[f'(x)dx\big]dx = \big[f''(x)dx\big]dx = f''(x)(dx)^2$$

Therefore (10.4) is equivalent to,

$$dy = 0 \text{ and } d^2y \leq 0 \tag{10.6}$$

Intuitively we are trying to maximize y so we look for a point at which dy is flat and in which in every direction y falls or at least does not rise. Now consider the function,

$$z = f(x, y)$$

Totally differentiating this function (by both variables), treating the $dx,\ dy$ terms[6] as constants and applying Young's theorem yields the following,

$$d^2z = \frac{\partial^2 f(x,y)}{\partial x^2}(dx)^2 + 2\frac{\partial^2 f(x,y)}{\partial x \partial y}(dxdy) + \frac{\partial^2 f(x,y)}{\partial y^2}(dy)^2 \tag{10.7}$$

A sufficient condition for a local maximum is,

$$d^2z \leq 0 \ \forall x, y \in D \tag{10.8}$$

That is, if the change on the slopes of all possible tangent planes drawn from the candidate solution (which has zero slope) are negative or flat the candidate solution is a local maximum. But how do we use this? Given that it must apply over all possible values of x and y we need *restrictions* on the partials that ensure (10.7) is less than or equal to zero.

Digression: Quadratic forms A quadratic form in n variables is of the form,

$$Q(x_1, x_2, \ldots, x_n) = \sum_{i,j=1}^{n} a'_{ij} x_i x_j \tag{10.9}$$

For 3 variables,

$$Q(\cdot) = a'_{11}x_1^2 + a'_{12}x_1x_2 + a'_{13}x_1x_3 + a'_{21}x_2x_1 + a'_{22}x_2^2 + a'_{23}x_2x_3 + a'_{31}x_3x_1 + a'_{32}x_3x_2 + a'_{33}x_3^2$$

Note that exponents in each term sum to 2 (There is a 1 on x_1 and x_2 in the second term etc) - and that we could have written this as, using that $a_{ij} = a'_{ii}$ and that $a_{ij} = a'_{ij} + a'_{ji}$.

$$Q(x_1, x_2, x_3) = a_{11}x_1^2 + a_{12}x_1x_2 + a_{13}x_1x_3 + +a_{22}x_2^2 + a_{23}x_2x_3 + a_{33}x_3^2$$

[6] We are free to choose the choice variables $x,\ y$ hence they can be treated as constants, we are not free to choose $f(x, y)$. If there is a constraint only one of the choice variables is free, once one is chosen the other will have to satisfy the constraint and this modifies the second order conditions.

In matrix language the general quadratic form is written as,

$$\mathbf{x}'\mathbf{Q}\mathbf{x} = [x_1 \dots x_n] \begin{bmatrix} a_{11} & .5a_{12} & \dots & .5a_{1n} \\ .5a_{12} & a_{22} & \dots & .5a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ .5a_{1n} & .5a_{2n} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \tag{10.10}$$

This matrix is symmetric.
Try writing the following using (10.10)

$$f(\mathbf{x}) = x^2 + 2x_2^2 + 3x_3^2 + 4x_1x_2 - 6x_1x_3 + 8x_2x_3$$

The shape of a 3 dimensional quadratic form

In 3 dimensions a quadratic form can only take 5 possible shapes, a hill, a hill with a flat top, a bowl, a bowl with a flat base and finally a saddle. The shape of the quadratic form depends entirely on the matrix $\mathbf{Q}$.

- If the function is zero only at $\mathbf{x}^*$ and positive everywhere else it must be a bowl, and it is called **positive definite**.
- If it is negative everywhere else it is a hill, and is called **negative definite**.
- The second order condition in (10.7) is a quadratic form so we need to see if it is positive or negative definite at the candidate solution $\mathbf{x}^*$.
- If at $\mathbf{x}^*$ (10.7) is positive (negative) definite it must also be a bowl (hill) with $\mathbf{x}^*$ as the minimum (maximum).
- The ***definiteness*** of the quadratic form depends entirely on the matrix $\mathbf{H}$ (Hessian matrix), and $\mathbf{H}$ is referred to as being positive definite (bowl), negative definite (hill), semi-positive definite (bowl with flat base), semi-negative definite (hill with a flat top) and indefinite (saddle).

Summary: The quadratic form

$$f(x, y) = ax^2 + 2bxy + cy^2$$

is:

$$\begin{aligned}
\text{positive definite} \quad &\Leftrightarrow \quad a > 0,\ \textit{and} \begin{vmatrix} a & b \\ c & d \end{vmatrix} > 0 \\
\text{positive semidefinite} \quad &\Leftrightarrow \quad a \geq 0,\ \textit{and} \begin{vmatrix} a & b \\ c & d \end{vmatrix} \geq 0 \\
\text{negative definite} \quad &\Leftrightarrow \quad a < 0,\ \textit{and} \begin{vmatrix} a & b \\ c & d \end{vmatrix} > 0 \\
\text{negative semidefinite} \quad &\Leftrightarrow \quad a \leq 0,\ \textit{and} \begin{vmatrix} a & b \\ c & d \end{vmatrix} \geq 0 \\
\text{indefinite} \quad &\Leftrightarrow \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix} < 0
\end{aligned}$$

Applied to the Second Order Condition

We can rewrite the second total differential (10.7), as the matrix ***quadratic form***,

$$d^2z = [dx\ dy] \begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix} = \mathbf{q}'\mathbf{D^2}f(\mathbf{x})\mathbf{q} = \mathbf{q}'\mathbf{H}\mathbf{q}\ \forall x, y \tag{10.11}$$

For a maximum $d^2z < 0$ for any values of $dx,\ dy$ that are assumed. This means that for $d^2z < 0$ the matrix **H must be negative definite** - so that dz is zero at (exactly at) the candidate solution and negative elsewhere. The relevant second order conditions apply to the matrix **H** alone. The ***ordered matrix*** of ***second partial derivatives*** is the **Hessian**. This argument extends in a logical manner to n variables.

8.2.2 Summary of Conditions for an Unconstrained Optimization Problem

Let

$$y = f(x_1, x_2, \ldots, x_n) = f(\mathbf{x})$$

A **Necessary** but not **Sufficient F.O.C** for a maxima or minima is,

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = 0\ \forall i \tag{10.12}$$

A **Sufficient S.O.C** for a local maximum/minimum is,

$$\mathbf{H}_{n\times n} = \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ f_{n1} & f_{n2} & \cdots & f_{nn} \end{bmatrix} \tag{10.13}$$

If $\mathbf{H}$ is **negative definite** then $\mathbf{x}^*$ is a **maximum**, whilst if $\mathbf{H}$ is **positive definite** $\mathbf{x}^*$ is a **minimum**. Along with the first order conditions these are sufficient for local maxima/minima. The semi-definite versions are necessary but not sufficient second order conditions. If $\mathbf{H}$ is *indefinite* $\mathbf{x}^*$ is a saddle point.

Digression: Testing for sign definiteness.

From $\mathbf{H}$ we can form the **leading principal minors** D_k, which are the determinants formed by deleting the last $n-k$ rows and the last $n-k$ columns. So if $k=1$ this means you delete all $n-1$ rows and columns giving the matrix with a number of rows and columns equal to $n-(n-1)=1$. For a 3×3 matrix if $k=1$ you delete the last 3-1 rows and columns leaving a 1×1 scalar. That is,

$$k=1 \Rightarrow D_1 = f_{11}$$

If $k=2$ the determinant of size $n-(n-2)=2$, that is formed by deleting the last $n-2$ rows and columns. This is,

$$k=2 \quad \Rightarrow \quad D_2 = \begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix}$$

$$k=3 \quad \Rightarrow \quad D_3 = \begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix}$$

A matrix of size n will have n such leading principal minors. An easy way to derive these leading principal minors is just to remember to form all the determinants you can starting from the 1×1 scalar in the top left corner, then the 2×2 from the top left corner and so on until you get the determinant of the whole matrix. Having done this we have,

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Second Order Conditions for Unconstrained Optimization

- $d^2z < 0 \Leftrightarrow \mathbf{H}$ is ***negative definite*** and x^* is a maximum if $(-1)^k D_k > 0$ for each and every k.

- $d^2z > 0 \Leftrightarrow$ **H** is ***positive definite*** and x^* is a minimum if $D_k > 0$ for each and every k.

  $\rightarrow$ So if **every** leading principal minor is positive x^* is a minimum, whilst if **every even** leading principal minor is positive and **every odd** leading principal minor is negative x^* is a maximum.

- $d^2z \leq 0 \Leftrightarrow$ **H** is ***negative semi-definite*** and $(-1)^k D_k \geq 0$ for each and every k. This is a necessary but not sufficient second order condition for x^* to be a maximum.

- $d^2z \geq 0 \Leftrightarrow$ **H** is ***positive semi-definite*** and $D_k \geq 0$ for each and every k. This is a necessary but not sufficient second order condition for x^* to be a minimum.

- If neither of the above is true then **H** is ***indefinite*** and x^* is neither a maximum nor a minimum.

Example: Find the maximum point(s) of,

$$f(x, y) = x^4 + x^2 - 6xy + 3y^2$$

Step 1: The F.O.C.

$$\begin{aligned} f_x &= 4x^3 + 2x - 6y = 0 \\ f_y &= -6x + 6y = 0 \end{aligned}$$

Step 2: Solve for candidate solutions x^*, y^*. From the second equation we get

$$x = y.$$

Replacing that into the first equation leads to

$$\begin{aligned} 4x^3 + 2x - 6x &= 0 \\ \Rightarrow 4x(x^2 - 1) &= 0 \end{aligned}$$

There are three solutions satisfying the above: $x = y = 0$, $x = y = 1$ and $x = y = -1$. Now apply the S.O.C.

Step 3: The S.O.C in Hessian form

$$\mathbf{H} = \begin{bmatrix} 12x^2+2 & -6 \\ -6 & 6 \end{bmatrix}$$

Step 4: Evaluate at each candidate point. For example if $x = y = 1$ we have,

$$\begin{aligned} D_1 &= 14 \\ D_2 &= \begin{vmatrix} 14 & -6 \\ -6 & 6 \end{vmatrix} \end{aligned}$$

This is a minimum - you check the other possibilities.

8.3 The Envelope Theorem

Theorem 9. *Let $f(\mathbf{x}; a)$ be a $\mathcal{C}^1$ function of $x \in \mathbb{R}^n$ and the scalar a. For each choice of the parameter a, consider the unconstrained maximization problem*

$$\max_{\mathbf{x}} f(\mathbf{x}; a).$$

Let $\mathbf{x}^*(a)$ be a solution of this problem. Suppose that $\mathbf{x}^*(a)$ is a $\mathcal{C}^1$ function of a. Then,

$$\frac{d}{da} f(\mathbf{x}^*(a); a) = \frac{\partial}{\partial a} f(\mathbf{x}^*(a); a) \tag{10.14}$$

Proof. We compute via the Chain Rule that

$$\begin{aligned} \frac{d}{da} f(\mathbf{x}^*(a); a) &= \underbrace{\frac{\partial}{\partial a} f(\mathbf{x}^*(a); a) \cdot 1}_{\text{Direct Effect}} + \underbrace{\sum_i \frac{\partial}{\partial x_i} f(\mathbf{x}^*(a); a) \cdot \frac{d}{da} x_i^*(a)}_{\text{Indirect Effect}} \\ &= \frac{\partial}{\partial a} f(\mathbf{x}^*(a); a) \end{aligned}$$

Since $\frac{\partial}{\partial x_i} f(\mathbf{x}^*(a); a) = 0$ for $i = 1, \dots, n$, by the usual first order conditions. □

9 Constrained Optimization (on 07/22)

Rarely in life are we confronted with an unconstrained optimization problem. The whole driving force and need for economics is *scarcity*. There are typically two types of constrained optimization discussed: with *equality* and with *inequality*. To see the motivation behind this, consider a utility function with a "Bliss Point" and a budget constraint.

However, we can approach this problem in a general fashion and use a *weak* inequality for the constraint. Thus the constrained maximization problem becomes:

$$\begin{aligned} \max_{\mathbf{x}} \quad & f(\mathbf{x}) \\ s.t. \quad & g(\mathbf{x}) \leq c \end{aligned}$$

where $g(\mathbf{x}) \leq c$ is the constraint and c is a constant. Note that we can just as easily make this a minimization problem by replacing max with min. The key to finding the *correct* x^* is to check our second order conditions. So how do we solve this problem? First, for intuition, let us work through the optimization problem when the constraint binds (or with equality).

9.1 Lagrange Multiplier Approach

The Lagrange multiplier method is a convenient way of solving an optimization problem

$$\begin{aligned} \max_{\mathbf{x}} \quad & f(\mathbf{x}) \\ s.t. \quad & g(\mathbf{x}) = c. \end{aligned}$$

To do so, follow these steps:
Step 1: Form what is called the ***Lagrangian Function***

$$L(\mathbf{x}) = f(\mathbf{x}) + \lambda[c - g(\mathbf{x})] \tag{11.1}$$

where $\lambda \in \mathbb{R}$ and is the "Lagrange Multiplier".

Step 2: Bring out the F.O.N.C. (first order necessary conditions)

$$\frac{\partial L(\mathbf{x})}{\partial x_i} = f_{x_i}(\mathbf{x}) - \lambda g_{x_i}(\mathbf{x}) = 0 \quad \forall i \in [1, n] \tag{11.2}$$

$$\frac{\partial L(\mathbf{x})}{\partial \lambda} = 0 \tag{11.3}$$

This gives $n+1$ conditions – the n choice variables and 1 for the constraint.

Step 3: Solve the above system for all the choice variables and λ

Step 4: Apply the S.O.C. – Derive

$$\frac{\partial L^2(\mathbf{x})}{\partial x_i x_j} = f_{x_i x_j}(\mathbf{x}) - \lambda g_{x_i x_j}(\mathbf{x}) \quad \forall i \in [1,n],\ j \in [1,n] \tag{11.4}$$

$$g_{x_i} \quad \forall i \in [1,n] \tag{11.5}$$

From these, you write out what's called the ***Bordered Hessian***

$$|\overline{\mathbf{H}}| = \begin{vmatrix} 0 & g_{x_1} & \cdots & g_{x_n} \\ g_{x_1} & f_{x_1x_1} - \lambda g_{x_1x_1} & \cdots & f_{x_1x_n} - \lambda g_{x_1x_n} \\ \vdots & \ddots & \ddots & \vdots \\ g_{x_n} & f_{x_1x_n} - \lambda g_{x_1x_n} & \cdots & f_{x_nx_n} - \lambda g_{x_nx_n} \end{vmatrix}$$

and again, if $\overline{\mathbf{H}}$ is ***positive definite*** then $\mathbf{x}^*$ is a *minimum* and if $\overline{\mathbf{H}}$ is ***negative definite*** then $\mathbf{x}^*$ is a *maximum*. Note that the requirements for definiteness are different with a bordered Hessian!!

Second Order Conditions for Constrained Optimization

- $d^2z < 0 \Leftrightarrow \mathbf{H}$ is ***negative definite*** and x^* is a maximum if $(-1)^k B_k > 0$ for each and every k.
- $d^2z > 0 \Leftrightarrow \mathbf{H}$ is ***positive definite*** and x^* is a minimum if $B_k < 0$ for each and every k.

  $\rightarrow$ So if **every** leading principal minor is negative x^* is a minimum, whilst if **every even** leading principal minor is positive and **every odd** leading principal minor is negative x^* is a maximum.

IMPORTANT: The principle minors for a bordered Hessian are different in that they include the border. For instance B_2 is a 3×3 matrix – think of taking the principle minor of the unconstrained Hessian and adding the border. See Appendix 11

Example: Suppose we have the following utility maximization problem with a budget constraint:

$$\begin{aligned} \max_{x,y} \quad & x^{\frac{1}{4}} y^{\frac{3}{4}} \\ s.t. \quad & 2x + 9y = 30 \end{aligned}$$

→ The Lagrangian is

$$L(x,y) = x^{\frac{1}{4}} y^{\frac{3}{4}} + \lambda[30 - 2x - 9y]$$

→ The F.O.N.C are:

$$\frac{\partial L(\cdot)}{\partial x} = \frac{1}{4}\left(\frac{y}{x}\right)^{\frac{3}{4}} - 2\lambda = 0 \tag{11.6}$$

$$\frac{\partial L(\cdot)}{\partial y} = \frac{3}{4}\left(\frac{x}{y}\right)^{\frac{1}{4}} - 9\lambda = 0 \tag{11.7}$$

$$\frac{\partial L(\cdot)}{\partial \lambda} = 30 - 2x - 9y = 0 \tag{11.8}$$

Now we need to find the values for x, y and λ. There are multiple approaches to this; one way is to solve for λ in (11.6) and (11.7), then divide one by the other.

$$\begin{aligned} \lambda &= \frac{1}{8}\left(\frac{y}{x}\right)^{\frac{3}{4}} \\ \lambda &= \frac{1}{12}\left(\frac{x}{y}\right)^{\frac{1}{4}} \\ \Rightarrow 1 &= \frac{\frac{1}{8}\left(\frac{y}{x}\right)^{\frac{3}{4}}}{\frac{1}{12}\left(\frac{x}{y}\right)^{\frac{1}{4}}} \\ \Rightarrow 1 &= \frac{3}{2}\left(\frac{y}{x}\right) \\ \therefore x &= \frac{3}{2}y \end{aligned}$$

Plugging this back into (11.8), we can solve for y

$$\begin{aligned} 30 &= 3y + 9y \\ 30 &= 12y \end{aligned}$$

$$\therefore y^* = \frac{5}{2} = 2.5 \quad \Rightarrow \quad x^* = \frac{15}{4} = 3.75$$

Furthermore, we can use these values to solve for λ

$$\lambda = \frac{1}{8}\left(\frac{2}{3}\right)^{\frac{3}{4}}$$

NOW, all we did was find the critical values for x and y, we need to check out S.O.C to determine we have, in fact, found a maximum!! First, derive the second derivatives

$$\frac{\partial^2 L(\cdot)}{\partial x^2} = \frac{-3}{16}\frac{y^{\frac{3}{4}}}{x^{\frac{7}{4}}} \tag{11.9}$$

$$\frac{\partial^2 L(\cdot)}{\partial xy} = \frac{3}{16}\frac{1}{x^{\frac{3}{4}}y^{\frac{1}{4}}} \tag{11.10}$$

$$\frac{\partial^2 L(\cdot)}{\partial y^2} = \frac{-3}{16}\frac{x^{\frac{1}{4}}}{y^{\frac{5}{4}}} \tag{11.11}$$

$$g_x = 2 \tag{11.12}$$

$$g_y = 9 \tag{11.13}$$

The Bordered Hessian is

$$|\overline{\mathbf{H}}| = \begin{vmatrix} 0 & 2 & 9 \\ 2 & \frac{-3}{16}\frac{y^{\frac{3}{4}}}{x^{\frac{7}{4}}} & \frac{3}{16}\frac{1}{x^{\frac{3}{4}}y^{\frac{1}{4}}} \\ 9 & \frac{3}{16}\frac{1}{x^{\frac{3}{4}}y^{\frac{1}{4}}} & \frac{-3}{16}\frac{x^{\frac{1}{4}}}{y^{\frac{5}{4}}} \end{vmatrix} = B_2$$

$$\begin{aligned} B_2 &= 36\frac{3}{16}\frac{1}{x^{\frac{3}{4}}y^{\frac{1}{4}}} - 81\left(\frac{-3}{16}\frac{y^{\frac{3}{4}}}{x^{\frac{7}{4}}}\right) - 4\left(\frac{-3}{16}\frac{x^{\frac{1}{4}}}{y^{\frac{5}{4}}}\right) \\ &= \frac{3}{16}\left[\frac{36}{x^{\frac{3}{4}}y^{\frac{1}{4}}} + 81\frac{y^{\frac{3}{4}}}{x^{\frac{7}{4}}} + 4\frac{x^{\frac{1}{4}}}{y^{\frac{5}{4}}}\right] > 0 \end{aligned}$$

Note that this inequality follows from the fact that both x and y are positive. Thus we do not have to explicitly solve for B_2 and we have confirmed that $\overline{\mathbf{H}}$ is negative definite and we have a found a maximum.

Earlier we solved for λ; but what exactly does this mean? After solving our three equations we have x^*, y^* if we substitute these optimal values into the objective function we obtain its maximum value subject to the constraint. This value is a function of the parameters of the model and is called the optimal or maximal value function,[7]

$$v = f(x^*, y^*) = f(x^*(c), y^*(c))$$

[7] You will later see this as the "Indirect Utility function" in your micro course.

Indeed as the values x^*, y^* must satisfy the constraint, the solutions should be function of the constant c. The parameter λ should also be a function of c as well. Totally differentiating the above expression yields

$$dv = f_{x^*}dx + f_{y^*}dy.$$

From F.O.N.C. we know that

$$\begin{aligned} f_{x^*} &= \lambda g_x \\ f_{y^*} &= \lambda g_y \\ \Rightarrow dv &= \lambda(g_x dx + g_y dy) \end{aligned}$$

Note that totally differentiating the constraint gives

$$dc = g_x dx + g_y dy$$

Therefore

$$\lambda = \frac{dv}{dc} \tag{11.14}$$

Which means the Lagrange Multiplier is the change in the optimal value function that is attributable to a small relaxation of the constraint. In other words, λ is the rate at which the (optimal) value of the objective function changes with respect to changes in the constraint constant c. This is also called the ***shadow price***.

In the utility example the consumer chooses the best vector of consumption goods given their prices and income. This yields the optimal level of utility subject to the constraint. It is a function of the prices, income and utility parameters. The Lagrange Multiplier reflects the increase in the optimal level of utility that occurs if the consumer has a unit more of income.

9.2 The Kuhn-Tucker Condition and Inequality Constraints

So far it is assumed the constraint binds. In general we do not know beforehand if the constraint binds or not. However, we do know if the constraint binds the solution satisfies the Lagrangian conditions, or if the solution does not bind it is an unconstrained problem. The problem is to find an additional condition that allows us to distinguish between these two possibilities. This condition is known as the complementary slackness condition.

9.2.1 The Basic Idea

Suppose we wish to maximize

$$f(x)\ s.t.\ g(x) \leq c$$

Treat the constraint as an equality and write out the Lagrangian,

$$L = f(x) + \lambda[c - g(x)]$$

The F.O.N.C. is

$$\frac{\partial L}{\partial x} = f'(x) - \lambda g'(x) = 0$$

This gives us an equation with two variables but we might be able to use the constraint with equality. There are two possibilities - at the solution the constraint either binds or it doesn't.

The constraint doesn't bind

If $g(x^*) < c$ the constraint has no effect on the choice of x^* so that the marginal value of relaxing the constraint must be zero,

$$g(x^*) < c \Leftrightarrow \lambda = 0$$

But if $\lambda = 0$ the F.0.N.C simplifies to that of an unconstrained problem, $f'(x) = 0$. There is now just one variable in 1 unknown.

The constraint does bind

By the same argument if,

$$\lambda > 0 \Leftrightarrow g(x^*) = c$$

This gives F.O.N.C. for the constrained case plus the constraint with equality - two equations with two unknowns. The extra bit the **Kuhn Tucker theorem** adds is that,

$$\lambda \geq 0$$

Or rather that,

$$\lambda[c - g(x)] = 0$$

This latter condition is called ***Complementary Slackness*** and it holds because either, $\lambda = 0$ or $g(x) = c$.

9.2.2 Lagrange plus Kuhn-Tucker: The Method

We first start with the simple case,

$$\max_{x,y} f(x,y)$$
$$s.t.\ g(x,y) \leq c$$

Recipe:

1. Associate a constant Lagrange multiplier λ with the constraint $g(x,y) \leq c$ and define

$$L(x,y) = f(x,y) + \lambda[c - g(x,y)]$$

2. Equate partial derivatives of L wrt x and y to zero

$$\begin{aligned} L_x &= f_x(x,y) - \lambda g_x(x,y) = 0 \\ L_y &= f_y(x,y) - \lambda g_y(x,y) = 0 \end{aligned}$$

3. Complementary Slackness Condition

$$\lambda[c - g(x,y)] = 0 \text{ with } \lambda \geq 0$$

4. Require (x,y) to satisfy the constraint $g(x,y) \leq c$.

5. Find all variables (x,y,λ) satisfying all of these conditions.

6. You need to compare the different solutions to see which one is the maximum.

We now go to the more general case.

We will assume many variables and many ***inequality*** constraints

$$\max_{\mathbf{x}} f(\mathbf{x})\ s.t.\ g_j(\mathbf{x}) \leq c_j \quad j = 1, \ldots, m$$

1. Set up the Lagrangian as before,

$$L(\mathbf{x}) = f(\mathbf{x}) + \sum_{j=1}^{m} \lambda_j [c_j - g_j(\mathbf{x})]$$

2. The F.O.C. are,

$$\frac{\partial L(\mathbf{x})}{\partial x_i} = \frac{\partial f(\mathbf{x})}{\partial x_i} - \sum_{j=1}^{m} \lambda_j \frac{\partial g_j(\mathbf{x})}{\partial x_i} = 0 \ \forall x_i$$

3. Now impose the complementary slackness condition,

$$\lambda_j \geq 0\big[= 0 \ if \ g_j(\mathbf{x}) < c_j\big] \ or \ \lambda_j[c_j - g_j(\mathbf{x})] = 0 \ with \ \lambda_j \geq 0 \ for \ j = 1, \ldots, m$$

   The complementary slackness ought to be intuitively clear. If the constraint doesn't bind the marginal value of relaxing the constraint must be zero, and hence the Lagrange multiplier is zero. On the other hand if the Lagrange multiplier is not zero the constraint must bind.

4. Require $\mathbf{x}$ to satisfy the constraints.

5. Find $\mathbf{x}$ and λ_j which satisfy all these constraints.

6. Compare the different solutions to see which is the maximum.

Example and Strategy:

Let us have a look at the following problem

$$\max_{x,y} f(x, y) = x^2 + y^2 - y \ s.t. \ x^2 + y^2 \leq 1$$

The best strategy is to just follow the logic of the K-T theorem. Set up the Lagrangian and take the F.O.C. Then,

1. Assume $\lambda = 0$ this will simplify to two F.O.C's. Solve these for x, y and then check to see if these values satisfy the constraint with strict inequality. If they do they form a candidate for a maximum.

2. Then assume $\lambda > 0$ so that $x^2 + y^2 = 1$. Now solve the three equations for the three unknowns - you will get an alternative candidate solution or the equations will be inconsistent. If you obtain an alternative solution then,

3. Compare the solutions by substituting into the objective function - the one with the highest value is the maximum.

Step 1: Form the Lagrangian:

$$L = x^2 + y^2 - y + \lambda[1 - x^2 - y^2]$$

Step 2: Find L_x, L_y and eliminate λ. The F.O.C. conditions are,

$$\begin{aligned} (1-\lambda)x &= 0 \\ (1-\lambda)2y &= 1 \\ \lambda[1 - x^2 - y^2] &= 0 \end{aligned}$$

1. Assume $\lambda = 0$, then from the first condition $x = 0$ and from the second condition $y = 1/2$. The constraint is satisfied with inequality so a candidate solution is,

$$x = 0, \lambda = 0, y = 0.5$$

2. Assume $x^2 + y^2 = 1$ and then along with the two F.O.C's solve them. Combining the two F.O.C. gives,

$$\begin{aligned} \frac{x}{2y} &= 0 \\ x^2 + y^2 &= 1 \end{aligned}$$

Which yields,

$$\begin{aligned} x &= 0 \\ y^2 = 1 \Rightarrow y &= \pm 1 \end{aligned}$$

Then from the second F.O.C. we have,

$$\lambda = \frac{1}{2} \quad y = -1 \Rightarrow \lambda = \frac{3}{2}$$

Giving two new candidate solutions,

$$x = 0, \lambda = \frac{1}{2}, y = 1 \quad x = 0, \lambda = \frac{3}{2}, y = -1$$

How do you check which is the optimal solution? Simply substitute into $f(x, y) = x^2 + y^2 - y$ to see which gives the largest value.

Example: Perfect Substitutes

Suppose we have the following maximization problem:

$$\max_{x,y} 2x + 4y$$
$$s.t.\ 3x + 2y \leq 20$$
$$-x \leq 0$$
$$-y \leq 0$$

Step 1: Set up the Lagrangian

$$L(x, y, \lambda) = 2x + 4y + \lambda_1[20 - 3x - 2y] + \lambda_2[0 + x] + \lambda_3[0 + y]$$

Step 2: The F.O.N.C. are

$$L_x = 2 - 3\lambda_1 + \lambda_2 = 0$$
$$L_y = 4 - 2\lambda_1 + \lambda_3 = 0$$

Step 3: Complementary Slackness

$$\lambda_1[20 - 3x - 2y] = 0\ s.t.\ \lambda_1 \geq 0$$
$$\lambda_2 x = 0\ s.t.\ \lambda_2 \geq 0$$
$$\lambda_3 y = 0\ s.t.\ \lambda_3 \geq 0$$

Step 4: Find the x, y and λs that satisfy these constraints. Suppose none of the constraints bind and $\lambda_1 = \lambda_2 = \lambda_3 = 0$, then

$$2 = 0$$
$$4 = 0$$

which cannot be. Suppose that $\lambda_1 > 0$ and $\lambda_2 = \lambda_3 = 0$, then

$$\lambda_1 = \frac{2}{3}$$
$$\lambda_1 = 2$$

which cannot be. Suppose that $\lambda_1 > 0$, $\lambda_2 > 0$, and $\lambda_3 = 0$, then

$$\begin{aligned} \lambda_1 &= \frac{2}{3} + \frac{1}{3}\lambda_2 \\ \lambda_1 &= 2 \\ 2y &= 20. \end{aligned}$$

Check the other case and see that this maximizes the function.

10 1^{st} Order Differential and Difference Equations (on 07/23)

10.1 Differential Equations

Here is a bit more explicit derivation of the solution to a first order differential equation. Suppose we have the following differential equation:

$$\dot{y} + Py = Q$$

1. Suppose $\dot{y} = 0$, then the "Particular Solution" (PS) is

$$y = \begin{cases} \frac{Q}{P} & if \ P \neq 0 \\ Qt & if \ P = 0 \end{cases}$$

2. Suppose $Q = 0$, then the "Complimentary function" (CF) is

$$\dot{y} = -Py \tag{12.1}$$

Now there's no real good way to explain this better than some sort of guess and verify type of thing.... Suppose

$$y = ce^{-Pt} \tag{12.2}$$

where c is a constant and e is the exponential function. Now differentiate (12.2) with respect to time t

$$\dot{y} = -Pce^{-Pt} \tag{12.3}$$

Now, recall equation (12.1) and use that with equation (12.3)

$$\begin{aligned}\dot{y} &= -Pce^{-Pt} = -Py \\ \therefore y &= ce^{-Pt}\end{aligned}$$

3. Now combine the two solutions to arrive at the General Solution (GS), which is y as a function of t

$$y(t) = \begin{cases} \frac{Q}{P} + ce^{-Pt} & if\ P \neq 0 \\ Qt + c & if\ P = 0 \end{cases}$$

4. In order to solve for c, we need to know the initial condition; i.e. what is $y(0)$? Once we know this we can solve for c.

10.1.1 Examples: F.O. differential equations

1. $\dot{y} + 6y = 7 \Rightarrow y(t) = \frac{7}{6} + ce^{-6t}$

2. $\dot{y} = 8 \Rightarrow y(t) = 8t + c$

3. $\dot{K} + \delta K = I \Rightarrow K(t) = \frac{I}{\delta} + ce^{-\delta t}$
   What is the unknown constant c? It depends on the initial condition! Suppose $K(0) = K_0 \Rightarrow K(t) = \frac{I}{\delta} + \left(\frac{I}{\delta} - K_0\right) e^{-\delta t}$

10.1.2 Dynamics

It all depends on the value of P:

- Converges if $P > 0$.
- Linear Trend if $P = 0$.
- Diverges if $P < 0$.

10.2 Difference Equations

With difference equations, time is not continuous, but the technique is very similar to solving differential equations. Suppose we have the following difference equation:

$$y_{t+1} + Py_t = Q$$

1. Suppose $y_{t+1} = y_t$, then the "Particular Solution" (PS) is

$$y = \begin{cases} \frac{Q}{1+P} & if\ P \neq -1 \\ Qt & if\ P = -1 \end{cases}$$

2. Suppose $Q = 0$, then the "Complimentary function" (CF) is

$$y_{t+1} = -Py_t \tag{12.4}$$

which means that

$$y(t) = c(-P)^t \tag{12.5}$$

3. Now combine the two solutions to arrive at the General Solution (GS), which is y as a function of t

$$y(t) = \begin{cases} \frac{Q}{1+P} + c(-P)^t & if\ P \neq -1 \\ Qt + c & if\ P = -1 \end{cases}$$

4. In order to solve for c, we need to know the initial condition; i.e. what is $y(0)$? Once we know this we can solve for c.

10.2.1 Examples: F.O. Difference Equations

1. $y_{t+1} - \frac{1}{2}y_t = 6 \Rightarrow y(t) = \frac{6}{1-\frac{1}{2}} + c\left(\frac{1}{2}\right)^t = 12 + \frac{c}{2^t}$
2. $y_{t+1} + \frac{1}{4}y_t = 5 \Rightarrow y(t) = \frac{5}{1+\frac{1}{4}} + c\left(\frac{-1}{4}\right)^t = 4 + c\left(\frac{-1}{4}\right)^t$
3. $y_{t+1} + y_t = 4 \Rightarrow y(t) = 2 + c(-1)^t$
4. Assume $(0 < d < 1)$: $K_{t+1} - (1-d)K_t = I \Rightarrow K(t) = \frac{I}{d} + c(1-d)^t$
   What is the unknown constant c? It depends on the initial condition! Suppose $K(0) = K_0 \Rightarrow K(t) = \frac{I}{d} + \left(\frac{I}{d} - K_0\right)(1-d)^t$

10.2.2 Dynamics

It all depends on the value of P:

- Explodes away from $Q/(1+P)$ if $P < -1$.
- Linear Trend if $P = -1$.
- Converges to $Q/(1+P)$ if $P \in (-1, 0)$.
- Silly if $P = 0$.

- Damped oscillations to $Q/(1+P)$ if $P \in (0,1)$.
- Perpetual zigzag if $P = 1$.
- Divergent oscillations if $P > 1$.

11 Appendix: Bordered and Non-Bordered Hessians

Notation can be a bit tricky for this. So, first, let's rewrite the conditions for a non-bordered Hessian:

Regular Hessian

Maximum – Negative Definite	Minimum – Positive Definite
$\lvert H_1\rvert < 0; \lvert H_2\rvert > 0;$ $\lvert H_3\rvert < 0; \ldots; (-1)^n \lvert H_n\rvert > 0\ \forall n$	$\lvert H_1\rvert, \lvert H_2\rvert, \ldots, \lvert H_n\rvert > 0\ \forall n$

This is true for when n refers to the dimension of the matrix; i.e. H_2 is a 2×2 matrix. The thing to really remember is that Positive Definite means that all the determinants have the same sign and are positive where Negative Definite, the sign alternates in a particular way.

Things get a little trickier with the Bordered Hessian, in particular what the bordered principal minors $|\bar{H}_n|$ are. Recall the ***Bordered Hessian***

$$|\overline{H}| = \begin{vmatrix} 0 & g_1 & \ldots & g_n \\ g_1 & f_{11} - \lambda g_{11} & \ldots & f_{1n} - \lambda g_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ g_n & f_{1n} - \lambda g_{1n} & \ldots & f_{nn} - \lambda g_{nn} \end{vmatrix}$$

So, the question becomes, how do you define $|\bar{H}_1|$? Most textbooks define it to be

$$|\overline{H}_1| = \begin{vmatrix} 0 & g_1 \\ g_1 & f_{11} - \lambda g_{11} \end{vmatrix},$$

which is a 2×2 matrix. Under this definition we can describe the conditions as such Therefore, the rule for negative definite is the same regardless of whether the Hessian is bordered – however, in the case of the bordered Hessian, n does not refer to the dimension of the matrix.

Bordered Hessian

Maximum – Negative Definite	Minimum – Positive Definite
$\lvert\bar{H}_1\rvert < 0; \lvert\bar{H}_2\rvert > 0;$ $\lvert\bar{H}_3\rvert < 0; \ldots; (-1)^n\lvert\bar{H}_n\rvert > 0 \; \forall n$	$\lvert\bar{H}_1\rvert, \lvert\bar{H}_2\rvert, \ldots, \lvert\bar{H}_n\rvert < 0 \; \forall n$

Another possible way of writing this would be to define d as the dimension of the principle minor such that

$$|\overline{H}_2| = \begin{vmatrix} 0 & g_1 \\ g_1 & f_{11} - \lambda g_{11} \end{vmatrix},$$

and not $|\overline{H}_1|$. Then we could write the conditions to be

Bordered Hessian

Maximum – Negative Definite	Minimum – Positive Definite
$\lvert\bar{H}_1\rvert = 0; \lvert\bar{H}_2\rvert < 0;$ $\lvert\bar{H}_3\rvert > 0; \ldots; (-1)^d\lvert\bar{H}_d\rvert < 0 \; \forall d > 1$	$\lvert\bar{H}_2\rvert, \lvert\bar{H}_3\rvert, \ldots, \lvert\bar{H}_d\rvert < 0 \; \forall d > 1$

Which is the opposite sign of the regular Hessian. Again, the key thing to remember is that Positive Definite means that all the determinants have the same sign and are negative where Negative Definite, the sign alternates in a particular way.

Example from Class

We are trying find the critical values of the constrained optimization problem and determine if it's a maximum or minimum:

$$\begin{aligned} U =& 4x_1^{\frac{1}{4}} x_2^{\frac{3}{4}} \\ & s.t. \; 2x_1 + 3x_2 = 12 \end{aligned}$$

The Lagrangian is

$$\mathcal{L} = 4x_1^{\frac{1}{4}} x_2^{\frac{3}{4}} + \lambda(12 - 2x_1 - 3x_2)$$

The F.O.N.Cs are

$$\mathcal{L}_1 = \left(\frac{x_2}{x_1}\right)^{\frac{3}{4}} - 2\lambda = 0 \tag{13.1}$$

$$\mathcal{L}_2 = 3\left(\frac{x_1}{x_2}\right)^{\frac{1}{4}} - 3\lambda = 0 \tag{13.2}$$

$$\mathcal{L}_\lambda = 12 - 2x_1 - 3x_2 = 0 \tag{13.3}$$

Using (1) and (2), we get

$$\frac{x_2}{3x_1} = \frac{2}{3} \tag{13.4}$$

which means that $x_2 = 2x_1$. Plugging this back into our constraint, we get $x_1^* = \frac{3}{2}$ and $x_2^* = 3$.

Now we need to check to see if this is a maximum or minimum. Our bordered Hessian is

$$|\overline{H}| = \begin{vmatrix} 0 & g_1 & g_2 \\ g_1 & f_{11} - \lambda g_{11} & f_{12} - \lambda g_{12} \\ g_2 & f_{12} - \lambda g_{12} & f_{22} - \lambda g_{22} \end{vmatrix} = \begin{vmatrix} 0 & 2 & 3 \\ 2 & \frac{-3}{4x_1}\left(\frac{x_2}{x_1}\right)^{\frac{3}{4}} & \frac{3}{4x_2}\left(\frac{x_2}{x_1}\right)^{\frac{3}{4}} \\ 3 & \frac{3}{4x_2}\left(\frac{x_2}{x_1}\right)^{\frac{3}{4}} & \frac{-3}{4x_2}\left(\frac{x_1}{x_2}\right)^{\frac{1}{4}} \end{vmatrix}$$

Plugging in our values for x_1^* and x_2^*, we get

$$|\overline{H}| = \begin{vmatrix} 0 & 2 & 3 \\ 2 & \frac{-1}{2}(2)^{\frac{3}{4}} & \frac{1}{4}(2)^{\frac{3}{4}} \\ 3 & \frac{1}{4}(2)^{\frac{3}{4}} & \frac{-1}{4}\left(\frac{1}{2}\right)^{\frac{1}{4}} \end{vmatrix}$$

Now we need to find the sign of the leading principle minors. Since it depends on how you define the subscript, I will leave these out

$$|0| = 0$$

$$\begin{vmatrix} 0 & 2 \\ 2 & \frac{-1}{2}(2)^{\frac{3}{4}} \end{vmatrix} = -4 < 0$$

$$\begin{vmatrix} 0 & 2 & 3 \\ 2 & \frac{-1}{2}(2)^{\frac{3}{4}} & \frac{1}{4}(2)^{\frac{3}{4}} \\ 3 & \frac{1}{4}(2)^{\frac{3}{4}} & \frac{-1}{4}\left(\frac{1}{2}\right)^{\frac{1}{4}} \end{vmatrix} = (-1)^2 * 0 + (-1)^3 * 2 \begin{vmatrix} 2 & \frac{1}{4}(2)^{\frac{3}{4}} \\ 3 & \frac{-1}{4}\left(\frac{1}{2}\right)^{\frac{1}{4}} \end{vmatrix} + (-1)^4 * 3 \begin{vmatrix} 2 & \frac{-1}{2}(2)^{\frac{3}{4}} \\ 3 & \frac{1}{4}(2)^{\frac{3}{4}} \end{vmatrix}$$

$$= -2\left[\frac{-1}{2}\left(\frac{1}{2}\right)^{\frac{1}{4}} - \frac{3}{4}(2)^{\frac{3}{4}}\right] + 3\left[\frac{1}{2}(2)^{\frac{3}{4}} + \frac{3}{2}(2)^{\frac{3}{4}}\right]$$

$$= \left(\frac{1}{2}\right)^{\frac{1}{4}} + \frac{3}{2}(2)^{\frac{3}{4}} + \frac{3}{2}(2)^{\frac{3}{4}} + \frac{9}{2}(2)^{\frac{3}{4}} = \frac{16}{(2)^{\frac{1}{4}}} > 0$$

Therefore, the determinants of our leading principle minors are alternating in the particular way that makes it negative definite and we have found a maximum.